

Clinical Study Protocol

Optimizing Reperfusion to Improve Outcomes and Neurologic Function (ORION): A Multicenter, Double-Blind, Placebo-Controlled, Randomized, Parallel-Group, Phase 2/3 Study to Evaluate the Efficacy and Safety of JX10 in Acute Ischemic Stroke with Late Presentations

Protocol Number: JX10002

Protocol Date/Version:

27 May 2025, Protocol Amendment 3 (v 2.2)

11 Nov 2024, Protocol Amendment 2 (v 2.1)

21 Oct 2024, Protocol Amendment 1 (v 2.0)

14 May 2024, Original Protocol (v 1.0)

Sponsor: Corxel Pharmaceuticals, Inc.

40 10th Ave, Floor 7

New York, NY 10014, USA

Regulatory Agency Identifier Number(s):

IND NUMBER: 160639

EU CT Number: 2024-519521-37

The study will be conducted in compliance with the protocol, ICH-GCP, and any applicable regulatory requirements.

The information contained in this document, particularly unpublished data, is the property or under control of Corxel Pharmaceuticals, Inc. and is provided to you in confidence as an investigator, potential investigator, or consultant, for review by you, your staff, and an applicable Institutional Review Board or Independent Ethics Committee. The information is only to be used by you in connection with authorized clinical studies of the investigational drug described in the protocol. You agree to not disclose any of the information to others without written authorization from Corxel Pharmaceuticals, Inc., except to the extent necessary to obtain informed consent from those persons to whom the drug may be administered.

SPONSOR INFORMATION

**Corxel Pharmaceuticals, Inc.
(USA)**
40 10th Ave, Floor 7
New York, NY 10014
USA

**Corxel Pharmaceuticals (Shanghai)
Co., Ltd.**
1225 West Nanjing Road
22 F, JC Plaza
Shanghai 200040
China

For urgent medical issues in which the study Medical Monitor should be contacted, please refer to the operations manual's Official Study Contact List for complete contact information.

Corxel may transfer any or all of its study-related responsibilities to a contract research organization and other third parties; however, Corxel retains overall accountability for these activities.

SPONSOR SIGNATURE PAGE

This protocol was approved by:

Edmond Chen



Signer Name: Edmond Chen
Signing Reason: I approve this document
Signing Time: 5/31/2025 | 8:06:23 AM CST
F29CEF74A25A45019C8CC5370FAD47BC

Edmond Chen, MD, FACC
Executive Vice President, Chief Medical Officer (US/EU)
Corxel Pharmaceuticals, Inc.

Date

TABLE OF CONTENTS

SPONSOR INFORMATION	2
SPONSOR SIGNATURE PAGE	3
LIST OF ABBREVIATIONS.....	9
1. KEY STUDY ELEMENTS	12
1.1. Synopsis	12
2. INTRODUCTION	31
2.1. Overview of AIS.....	31
2.1.1. Pathophysiological Mechanism of Ischemic Stroke.....	31
2.1.2. Risk Factors for Stroke	32
2.1.3. Current Therapies for AIS	32
2.2. Profile of Previous Experience with JX10.....	33
2.2.1. Nonclinical Experience.....	33
2.2.2. Clinical Experience.....	34
2.3. Study Rationale.....	36
2.4. Rationale for Dosing Regimen	37
2.5. Benefit-Risk Assessment	38
3. STUDY OBJECTIVES AND ENDPOINTS.....	39
4. STUDY DESIGN	42
4.1. Study Overview	42
4.1.1. Neuroimaging Timing and Procedures	43
4.1.2. Assessments and Evaluation of Neuroimaging	44
4.2. Study Duration for Participants	44
4.3. Study Stopping Rules	45
4.4. Unscheduled Visits	45
4.5. End of Study	45
5. STUDY POPULATION	45
5.1. Inclusion Criteria	46
5.2. Exclusion Criteria	47
5.3. Screening, Retesting, and Screen Failures.....	50

5.3.1.	Screening	50
5.3.2.	Retesting	51
5.3.3.	Screen Failures.....	51
5.4.	Randomization.....	51
6.	STUDY TREATMENT.....	52
6.1.	Regimen.....	52
6.1.1.	Modification of Dose and/or Treatment Schedule.....	52
6.2.	Study Treatment Management.....	52
6.2.1.	Clinical Formulation.....	53
6.2.2.	Stability, Storage, and Handling Instructions	53
6.3.	Blinding Procedures.....	53
6.4.	Guidance on Care of Participants: Precautions, Monitoring, and Treatment	54
6.4.1.	General Considerations.....	54
6.4.2.	Blood Pressure and Neurological Monitoring	54
6.4.3.	Intracranial Hemorrhage or Suspected Intracranial Hemorrhage	55
6.4.4.	Clinically Significant or Major Bleeding	55
6.4.5.	Anaphylaxis and Hypersensitivity Reactions	55
6.5.	Compliance	55
6.6.	Concomitant Therapy and Procedures.....	56
6.6.1.	Concomitant Therapy	56
6.6.2.	Prohibited Concomitant Therapy.....	56
6.6.3.	Concomitant Procedures	56
6.7.	Continuation of Treatment.....	56
7.	DISCONTINUATION AND WITHDRAWAL OF PARTICIPANTS FROM THE STUDY	57
7.1.	Discontinuation of Study Treatment.....	57
7.2.	Lost to Follow-Up.....	57
7.3.	Withdrawal of Participants from the Study	58
8.	TRIAL ASSESSMENTS AND PROCEDURES	59
8.1.	Efficacy, Pharmacokinetic, and Pharmacodynamic Assessments	59
8.1.1.	Clinical Efficacy Assessments.....	59

8.1.2.	Laboratory Target Engagement and Pharmacodynamic Assessments	60
8.1.3.	Pharmacokinetic Assessments	60
8.1.4.	Pharmacodynamic Assessments	61
8.1.5.	Pharmacogenetic and Genetic Assessments	61
8.2.	Safety Assessments.....	61
8.2.1.	Clinical Safety Assessments	61
8.2.2.	Laboratory Safety Assessments	62
8.2.3.	Product-Specific Safety Assessments.....	63
8.3.	Safety Definitions, Recording, Reporting, And Responsibilities	63
8.3.1.	Definitions	63
8.3.2.	Safety Classifications.....	65
8.3.3.	Monitoring and Recording Events.....	67
8.3.4.	Procedures for Handling Special Situations	70
8.3.5.	Contraception Requirements	72
8.3.6.	Safety Responsibilities.....	74
9.	STATISTICAL METHODS AND DETERMINATION OF SAMPLE SIZE.....	75
9.1.	General Considerations.....	75
9.2.	Statistical Hypotheses	76
9.3.	Sample Size Considerations	76
9.4.	Multiplicity Adjustment.....	76
9.5.	Analysis Sets.....	77
9.5.1.	All Randomized Set.....	77
9.5.2.	Full Analysis Set.....	77
9.5.3.	Safety Analysis Set	77
9.5.4.	Pharmacokinetic/Pharmacodynamic Analysis Set	77
9.5.5.	Concentration-QTc Analysis Set	77
9.6.	Statistical Analysis.....	78
9.6.1.	Analysis for Efficacy Endpoints	78
9.6.2.	Analysis for Pharmacokinetic Endpoints.....	83
9.6.3.	Analysis for Pharmacodynamic Endpoints.....	83
9.6.4.	Analysis for Safety Endpoints	83

9.6.5.	Patient Disposition.....	85
9.6.6.	Demographics and Other Baseline Characteristics.....	85
9.6.7.	Concomitant Medications	85
9.6.8.	Interim Analysis.....	85
9.7.	Subgroup Analysis.....	86
10.	ETHICAL AND REGULATORY REQUIREMENTS.....	86
10.1.	Declaration of Helsinki.....	87
10.2.	Ethics Committee.....	87
10.3.	Changes to Final Protocol.....	87
10.4.	Informed Consent	88
10.5.	Participant Data Protection	89
10.6.	Compensation for Injury	89
10.7.	Conflict of Interest	89
10.8.	Study Report Signatory.....	89
10.9.	Registration of Study and Disclosure of Study Results.....	89
10.10.	Retention of Study Data.....	90
11.	KEY ROLES AND STUDY COMMITTEES	90
11.1.	Site Staff	90
11.2.	Vendors.....	91
11.2.1.	Contract Research Organization	91
11.2.2.	Interactive Response Technology.....	91
11.2.3.	Electronic or Remote Data Capture.....	91
11.2.4.	Bioanalytical Laboratory for Laboratory Assessments	91
11.2.5.	Central Facility for Other Assessments	91
11.2.6.	Central Review of Raters.....	91
11.3.	Study Committees.....	92
11.3.1.	Independent Data Monitoring Committee	92
12.	ADMINISTRATIVE PROCEDURES	92
12.1.	Study Site Initiation	92
12.2.	Quality Control and Quality Assurance.....	92
12.3.	Monitoring of the Study.....	93

12.4.	Study Funding.....	93
12.5.	Publications.....	93
13.	REFERENCES	94
14.	SIGNED AGREEMENT OF THE STUDY PROTOCOL.....	97

LIST OF TABLES

Table 1:	Schedule of Activities (for Part 1 and Part 2).....	26
Table 2:	Objectives and Endpoints	39
Table 3:	Estimand of Efficacy Endpoint Analysis.....	80

LIST OF FIGURES

Figure 1:	Schematic of Study Design.....	25
Figure 2:	mRS outcome on Day 90.....	35
Figure 3:	Key Results.....	36

LIST OF ABBREVIATIONS

ACA	anterior cerebral artery
AE	adverse event
AESI	adverse events of special interest
AIS	acute ischemic stroke
AOL	arterial occlusive lesion
aPTT	activated partial thromboplastin time
ASPECTS	Alberta Stroke Program Early CT Score
AUC	area under the curve
AUC ₀₋₂₄	area under the curve from time 0 to 24 hours after the start of dosing
BIS	Barthel Index Score
BP	blood pressure
C-QTcS	Concentration-QTc Analysis Set
CT	computed tomography
CTA	computed tomography angiography
CTP	computed tomography perfusion
CTP-CBF	CT perfusion-cerebral blood flow
DBP	diastolic blood pressure
DHET	dihydroxyeicosatrienoic acid
DSA	digital subtraction angiography
DWI	diffusion weighted imaging
ECG	electrocardiogram
eCRF	electronic case report form
EET	epoxyeicosatrienoic acid
eGFR	estimated glomerular filtration rate
EQ-5D-5L	EuroQol 5-level assessment of health outcomes
EVT	endovascular thrombectomy
EU	European Union
FAS	Full Analysis Set
FLAIR	Fluid-Attenuated Inversion Recovery
GCP	Good Clinical Practice

GDPR	General Data Protection Regulation
HCO-60	polyoxyethylene hydrogenated castor oil 60
IB	Investigator's Brochure
ICA	internal carotid artery
ICEs	intercurrent events
ICF	informed consent form
ICH	intracranial hemorrhage
IDMC	independent data monitoring committee
INR	international normalized ratio
IRT	interactive-response technology
IV	intravenous(ly)
LAR	legally authorized representative
LFT	liver function test
LKW	Last known well
LVO	large vessel occlusion
MAR	missing at random
MCA	middle cerebral artery
mCTA	multiphase CT angiography
MedDRA	Medical Dictionary for Regulatory Activities
mL	milliliter
MR	magnetic resonance
MRA	magnetic resonance angiography
MRI	magnetic resonance imaging
MRP	MR perfusion
mRS	Modified Rankin Scale
ms	millisecond
mTICI	modified thrombolysis in cerebral infarction (scale)
NCCT	non-contrast computed tomography
NIHSS	National Institutes of Health Stroke Scale
NINDS	National Institute of Neurological Disorders and Stroke
PAP	plasmin antiplasmin complex

PCA	posterior cerebral artery
PD	pharmacodynamics
PK	pharmacokinetics
PKPDS	Pharmacokinetics/pharmacodynamics dataset
PMDA	Pharmaceuticals and Medical Devices Agency (Regulatory Authority of Japan)
PT	prothrombin time
PWI	perfusion-weighted imaging
QTc	corrected QT interval
rtPA	recombinant tissue plasminogen activator
SAE	serious adverse event
SAP	statistical analysis plan
SBP	systolic blood pressure
sEH	soluble epoxide hydrolase
sICH	Symptomatic intracranial hemorrhage
SIS-16	Stroke Impact Scale - 16
SOC	system organ class
SUSAR	suspected unexpected serious adverse reaction
T2	transverse relaxation time
T _{max}	Time-to-maximum of the residue function
TIMELESS	Thrombolysis in Imaging-Eligible, Late-Window Patients to Assess the Efficacy and Safety of Tenecteplase
TOAST	Trial of Org 10172 in Acute Stroke Treatment
tPA	tissue plasminogen activator
UGT1A1	uridine diphosphate glucuronosyltransferase 1A1 gene
US	United States

1. KEY STUDY ELEMENTS

1.1. Synopsis

Protocol Title:	Optimizing Reperfusion to Improve Outcomes and Neurologic Function (ORION): A Multicenter, Double-Blind, Placebo-Controlled, Randomized, Parallel-Group, Phase 2/3 Study to Evaluate the Efficacy and Safety of JX10 in Acute Ischemic Stroke with Late Presentations
Protocol Number:	JX10002
Version Number:	2.2
Investigational Product	JX10 (formerly BIIB131/TMS-007)
Study Phase:	2/3
Study Indication:	Acute ischemic stroke
Study Rationale:	Acute ischemic stroke (AIS) is the second leading cause of death and disability worldwide. Mainstay of pharmacologic treatment is primary reperfusion with thrombolysis, which is limited to a narrow therapeutic time window. Thrombolysis with existing therapies has limitations in rate of recanalization and carries risk of excess intracranial hemorrhage. Due to a host of contributing factors beyond a patient's control, a majority of patients do not present for care in time, within 4.5 hours from the time when they were Last Known Well (LKW) to receive urgent treatment. There remains an unmet need in the treatment of AIS for safer and more effective thrombolytics, which can also be administered to a broader population with an extended treatment window. JX10 is a thrombolytic agent with a novel mechanism of action of inducing conformational changes in plasminogen to increase downstream fibrin affinity, as opposed to the systemic plasminogen activation characteristic of alteplase and tenecteplase. The resultant activity is a thrombosis-specific physiological fibrinolysis. In addition to this effect, JX10 has anti-inflammatory properties based on inhibition of soluble epoxide hydrolase (sEH) as well as antioxidant activity based on its chemical structure, which may suppress hemorrhagic changes associated with cerebral infarction. The anti-inflammatory effects of JX10 are thought to reduce the risk of intracranial hemorrhage associated with such treatment, via a reduction of endothelial cell damage at the site of thrombosis and preservation of vascular integrity.

Study Objectives and Endpoints:

Primary Efficacy Objective	Primary Efficacy Endpoints
To determine if JX10 improves functional outcome as measured by the mRS when compared with placebo following AIS	Proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days
Secondary Efficacy Objective	Secondary Efficacy Endpoints
To evaluate the effects of JX10 on acute and 90-day clinical outcomes	Ordinal mRS score, based on a 6-point ordinal scale at 90 days Proportion of participants with functional independence (mRS score 0-2) at 90 days Proportion of participants with BIS > 90 at 90 days Proportion of participants with favorable outcome (mRS 0-1 or 0-2), taking into account baseline NIHSS score (mRS sliding dichotomy) Proportion of participants with reperfusion at 24 hours post-treatment, defined as > 90% reduction in $T_{max} > 6$ seconds lesion volume or complete recanalization on CT angiography (CTA)/magnetic resonance angiography (MRA) Proportion of participants with recanalization at 24 hours post-treatment, defined as complete or partial recanalization on CTA/MRA Proportion of participants with angiographic reperfusion (TICI 2b/3 or TICI 3) at completion of angiographic procedure (endovascular thrombectomy participants only) Proportion of participants where planned thrombectomy was not performed (planned endovascular thrombectomy participants only) Proportion of participants with improvement on the NIHSS by > 7 points or score 0 or 1 at 7 days or hospital discharge

Primary Safety Objective	Primary Safety Endpoints
To evaluate the risk of symptomatic intracranial hemorrhage of JX10 in participants with acute ischemic stroke	Incidence of symptomatic intracranial hemorrhage, defined as local or remote parenchymal hemorrhage type 2, subarachnoid hemorrhage, and/or intraventricular hemorrhage within 36 hours post-randomization, combined with a neurological deterioration of 4 points or more on the NIHSS from baseline (the closest collection before administration of the study treatment), or from the lowest NIHSS value between baseline and 24 hours, or leading to death
Secondary Safety Objective	Secondary Safety Endpoints
To evaluate the safety and tolerability of JX10 in participants with acute ischemic stroke	Incidence of adverse events (AEs) and serious adverse events (SAEs) Incidence of any intracranial hemorrhage within 24 hours of study treatment Incidence of any intracranial hemorrhage within 14 days of study treatment Incidence of major bleeding within 24 hours of study treatment Incidence of major bleeding within 14 days of study treatment Incidence of symptomatic cerebral edema within 14 days of study treatment Proportion of participants with parenchymal hematoma type 2 within 7 days following treatment based on Heidelberg bleeding classification Proportion of participants with parenchymal hematoma type 1 or 2 within 7 days following treatment based on Heidelberg bleeding classification Overall mortality rates at 30 and 90 days

Study Design:

This is a randomized, multicenter, double-blind, placebo-controlled study designed to maximize operational efficiency modeled after established studies like the National Institute of Neurological Disorders and Stroke (NINDS) trial to dose-find and confirm the efficacy and safety of that dose in an appropriately sized study to support eventual regulatory registration of JX10. The study is intended to test whether a novel thrombolytic with anti-inflammatory actions would benefit a broad group of patients with AIS who present for care, within 4.5 to 24 hours. The enrollment and participation in the trial will not preclude or delay standard of care treatment including EVT.

The study will consist of 2 parts:

Part 1 will be a dose-ranging study to identify the optimal dose of JX10 based on assessment of safety and efficacy. A subset of participants with qualified imaging studies will be evaluated for angiographic revascularization. To establish the optimal dose in Part 1 of the study, participants will be randomized 1:1:1 to receive JX10 at dosing levels of 1 mg/kg, 3 mg/kg, or placebo, with approximately 80 participants in each arm. The primary efficacy outcome measure will be mRS at Day 90.

An interim analysis will be performed when the last participant in Part 1 has been enrolled. Based on best available data at the time of the interim analysis, a determination will be made by the independent data monitoring committee (IDMC) as to whether there is a dose that will be advanced to Part 2. The optimal dose of JX10 for Part 2 will be selected based on the totality of data, including radiographic, clinical, safety, and PK data from Part 1. The most efficacious dose with an acceptable safety profile will be selected as the dose to advance to Part 2. The acceptance of the recommendation will trigger the start of enrollment for Part 2, with site notification. Should no dosing arm demonstrate superiority relative to the placebo group, the Sponsor may decide to delay the start of Part 2 or terminate the study.

Part 1, upon the last participant completing the last visit, will close out, unblind, and report results when the last participant has completed the final visit. Regulatory interactions will proceed as needed. Select centers may have delayed start of enrollment into Part 2, pending additional approval or authorization by responsible regulatory authorities. Part 2 may be amended based on findings of Part 1 and requirements from respective regulatory authorities. Part 1 participants will not contribute to the analysis of Part 2. However, pertinent findings and learnings from Part 1 will be used to inform and potentially modify Part 2, including but not limited to modifications of endpoints, procedures, eligibility criteria, and sample size.

In Part 2 of the study, approximately 500 participants may be randomized 1:1 to receive either JX10 (optimal dose selected after the end of Part 1 analysis) or placebo. Enrollment of Part 2 will start, where permitted, while Part 1 completes the follow-up and database lock. The primary efficacy measure for Part 2 of the study is the same mRS outcome at Day 90. The primary analysis for Part 2 will include only participants enrolled in Part 2 of the study.

Participants who meet all eligibility criteria and provide a signed informed consent will be randomized to the JX10 or placebo arms. Participants are to receive the study drug administration as soon as possible with the prespecified treatment window, within 6 hours from LKW among participants with no perfusion imaging or within 24 hours with qualifying penumbral volume on perfusion or eligible imaging modality.

Participants will be monitored closely, especially in the first 24 hours, during which frequent blood pressure and neurological assessments will occur. Usual precautions should be exercised in participants who will undergo possible thrombectomy at least 1 hour following the initiation of study drug administration. Participants are followed for at least 90 days. Outcome assessments such as mRS or NIHSS should be performed by trained and certified professionals. Every effort should be made to complete all protocol-mandated procedures and evaluations. In recognition of the potential need for extended hospital stay, rehabilitation, mobility, and transportation issues, telehealth options are permitted.

Study Location: Approximately 150+ sites are planned globally.

Study Population: This study will be conducted in participants who meet the following criteria:

Inclusion Criteria

1. Age ≥ 18 and ≤ 90 years old (Age > 85 must be mRS 0 at baseline (premorbid) to be eligible).
2. Written informed consent by patient, his/her legally authorized representative, or independent physician where local regulation allows (in accordance with all local and national regulations or according to the local ethics committee's guidelines or by another process compliant with applicable national laws and regulations and ethics committee requirements).
3. Acute ischemic stroke with compatible clinical presentation and symptomatic high grade or complete occlusion of the intracranial internal carotid, M1, M2, or distal branches of the middle cerebral artery (MCA), anterior cerebral artery (ACA), or posterior cerebral artery (PCA), demonstrated by:

- a. Computer Tomography Angiography (CTA) or Magnetic Resonance Angiography (MRA) with high grade stenosis or occlusion of an eligible artery, or
 - b. Computer Tomography or Magnetic Resonance Perfusion (CTP/MRP) with focal deficit in an eligible artery, or
 - c. Non-contrast Computer Tomography (NCCT) with hyperdense artery sign in an eligible artery or Magnetic Resonance Imaging (MRI) with susceptibility vessel sign (defined as a hypointense signal exceeding the diameter of the contralateral artery at the thrombus site) in an eligible artery.
4. Radiographic evidence of salvageable tissue* is present based on:
- a. A mismatch ratio (perfusion lesion ($T_{\max} > 6$ seconds) volume/core estimate) > 1.2 ,
 - b. Estimated core infarct volume < 70 mL, and
 - c. Total mismatch volume ≥ 10 mL on CTP or magnetic resonance (MR) perfusion-weighted imaging (PWI)/MRI diffusion imaging.

The perfusion criteria apply to all participants, including those who present within 4.5 to 6 hours of LKW, if perfusion imaging is obtained. Perfusion is mandatory for patients presenting beyond 6 hours, optional for patients < 6 hours.

**The quantitative assessment of penumbra may include alternative methods (e.g., if the software uses different parameters in place of $T_{\max}$).*

5. Presentation with intended study treatment within 4.5 to 6 hours of Last Known Well (LKW) in the absence of radiographic assessment of perfusion or within 4.5 to 24 hours of LKW with radiographic evidence of penumbra meeting criterion (4).
6. Pre-treatment score of NIHSS ≥ 5 .
7. Functionally independent prior to stroke onset as evidenced by premorbid mRS < 2 (< 1 , if age 86-90).
8. All women of childbearing potential and all men must practice contraception as described in Section 8.3.5. For women of childbearing potential, a negative pregnancy test at screening is required. All women of childbearing potential must practice effective contraception for at least 30 days after their last dose of study treatment. All men must practice effective contraception during the study and for 90 days after their last dose of study treatment. In addition, participants should not donate sperm or eggs

during the study and for at least 90 days after their dose of study treatment.

Exclusion Criteria

Neurovascular Imaging

1. Radiographic findings pre-randomization of any of the following:
 - a. Large core infarction, evidenced by a core infarct volume > 70 mL, assessed on DWI or CTP; or extensive early ischemic change (hypodensity) on non-contrast CT estimated to be > 1/3 MCA territory, or significant hypodensity outside the $T_{max} > 6$ seconds perfusion lesion that invalidates mismatch criteria, or
 - b. Occlusion in more than 1 vascular territory confirmed on CTA/MRA, or
 - c. Significant mass effect or clinically significant cerebral edema per Investigator's judgement, or
 - d. Evidence of acute intracranial or extracranial hemorrhage, intracranial tumor (except small meningioma), neoplasm, or arteriovenous malformation), or
 - e. Clinical history, past imaging, or clinical judgement suggests that the intracranial occlusion is chronic.

Etiology of Neurologic Symptoms

2. Non-ischemic or non-thrombotic etiology of current stroke symptoms such as suspected cerebral vasospasm, infectious source (e.g., bacterial endocarditis, septic shock), complications from acute drug abuse (e.g., cocaine intoxication).

Timing of Endovascular Thrombectomy (EVT)

3. Availability of endovascular thrombectomy (EVT) for current AIS (in whom EVT is indicated) where planned EVT would be feasible (e.g., time of femoral artery puncture), within 1 hour following study drug infusion.

Medical History and Current Health Status

Bleeding Risks

4. Medical history or active clinically significant bleeding, lesions, or conditions (at the investigator's judgement) considered to be of significant risk for major bleeding; (this may include but not limited to any history of intracranial hemorrhage or vascular aneurysm of the large arteries of major intraspinal or intracerebral abnormalities).

5. Cerebral infarction within 90 days of screening.
6. Arterial dissection involving any intracranial artery or the aortic arch.
7. Confirmed acute coronary syndrome within 90 days of screening.
8. Severe hepatic impairment as defined by decompensated liver disease (e.g., Child-Pugh Class C) or clinically significant hepatic condition associated with coagulopathy or bleeding risk.
9. Medical history of chronic renal failure, any condition requiring dialysis or renal replacement therapy or evidence of acute kidney injury at the time of screening, an estimated glomerular filtration rate $< 60 \text{ mL/min/1.73m}^2$.
10. Severe, uncontrolled hypertension (systolic blood pressure $\geq 185 \text{ mmHg}$ or diastolic blood pressure $\geq 110 \text{ mmHg}$) that cannot be controlled with antihypertensive therapy.
11. Known bleeding diathesis (hereditary or acquired) or any significant coagulopathy. Specifically, platelet count $< 100,000/\mu\text{L}$, international normalized ratio > 1.7 , aPTT > 40 seconds, or prothrombin time > 15 seconds.
12. Major trauma, surgery, or invasive procedures:
 - a. Severe head trauma within 3 months of screening or acute head trauma.
 - b. Major trauma not involving the head within 14 days of screening.
 - c. Major surgery with 14 days of screening.
 - d. Intracranial or intraspinal surgery within 90 days of screening.
 - e. Dural puncture (e.g., lumbar puncture) or arterial puncture of a noncompressible blood vessel within 7 days of screening.

General

13. Pre-existing medical, neurological, or psychiatric disease that would confound the neurological or functional evaluations of this study.
14. Pre-treatment blood glucose $> 400 \text{ mg/dL}$ (22.20 mmol/L) or Pre-treatment blood glucose $< 50 \text{ mg/dL}$ (2.78 mmol/L) unless it is corrected prior to study treatment administration. Participants with subsequently normalized blood glucose levels may be considered for inclusion, per Investigator judgement.

15. Known hereditary or acquired abnormality of UGT1A1 metabolism or deficiency such as Gilbert's syndrome (hereditary liver condition with elevated bilirubin), Crigler-Najjar syndrome (UGT1A1 gene defect associated with congenital non-hemolytic jaundice).
16. Prolonged QT with QTc of > 450 ms for males and > 460 ms for females.
17. Life expectancy less than 6 months due to comorbid condition.

Medications

18. Prior thrombolytic administration within 90 days of screening or planned thrombolytic administration for the AIS. (Co-administration with any other thrombolytic agent(s) within 48 hours is prohibited).
19. Known or suspected use of any oral anticoagulant therapy, including but not limited to vitamin K antagonists, thrombin inhibitors, or factor Xa inhibitors, within 48 hours of screening unless blood testing confirms absence of drug substance in the system.
20. Use of dual anti-platelet therapy, IV aspirin, or glycoprotein IIb/IIIa inhibitors within 24 hours of screening. Pre-screening use of oral antiplatelet monotherapy with aspirin at doses less than or equal to 325 mg daily OR clopidogrel 75 mg daily is permitted.
21. Use of heparin or low molecular weight heparin at a therapeutic dose, excluding prescreening prophylactic dose low molecular weight heparin with a normal aPTT.
22. Use of nephrotoxic medications or agents within 7 days of screening that would (in the investigator's opinion) pose significant risk of acute kidney injury (e.g., aminoglycosides, cyclosporins, lactams, amphotericin B, cisplatin, indomethacin, etc.).

Other Exclusion Criteria

23. Prior treatment with JX10, any known history of systemic hypersensitivity reaction or anaphylaxis to JX10, the excipients contained in the formulation, and if applicable, any diagnostic agents anticipated to be administered during the study.
24. Participants who are pregnant or currently breastfeeding, and those intending to become pregnant during the study.

25. Current enrollment or a plan to enroll in another investigational device study, or interventional clinical study in which an investigational treatment or approved therapy for investigational use is administered within 90 days prior to the screening.
26. Substance abuse history including heavy alcohol use over the past year, per Investigator judgement, or acute intoxication with alcohol or other substance that may impair participant's ability to comply with treatment and assessment.
27. Inability to comply with study requirements or other unspecified reasons that, in the opinion of the Investigator or Sponsor, make the participant unsuitable for enrollment.

Number of Planned Participants:

Approximately 740 participants will be enrolled, including approximately 240 participants in Part 1 and approximately 500 participants in Part 2.

Treatment Groups:

A single dose of JX10 or placebo will be administered over 31-minutes (an IV bolus of 10% of the total dose over 1 minute, followed by a continuous IV infusion of the remaining 90% of the total dose over 30 minutes). Treatment arms in Part 1 and Part 2 are as follows:

Part 1 comprises the following treatment arms:

- 1 mg/kg JX10 (n = 80)
- 3 mg/kg JX10 (n = 80)
- Placebo (n = 80)

Part 2 comprises 2 treatment arms:

- Optimal dose of JX10 from Part 1 (n = 250)
- Placebo (n = 250)

Statistical Consideration:

Sample Size Determination:

In Part 1, each JX10 dose group (1 mg/kg, 3 mg/kg) will be compared to the placebo group for proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days. Odds ratio of improvement on proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days comparing each of the JX10 dose groups to the placebo group will be estimated. An odds ratio greater than 1 means JX10 is favored over placebo for this outcome. At a 1-sided significance level of 0.05, 80 participants per arm will provide more than 90% power when the true odds ratio is 3.0 for a JX10 dose group compared to the placebo group and when the proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days is assumed to be 18.4% for placebo (based on observations from the Japanese Phase 2a clinical experience). No multiplicity adjustment will be made.

In Part 2, the odds ratio of improvement on proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days comparing JX10 to the placebo will be estimated. An odds ratio greater than 1 means JX10 is favored over placebo for this outcome. Assuming the odds ratio of JX10 to placebo is 2.0, the proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days in placebo group is assumed to be 18.4%, the 2-sided type I error rate as 0.05, a sample size of 500 (approximately 250 randomized to JX10 and 250 randomized to placebo) provides 91% power to detect the difference in proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days between JX10 and placebo. The sample size of Part 2 might be modified based on the interim analysis result of Part 1 if there is significant deviation between the actual interim analysis result from the estimated parameter values to maintain the intended power.

Estimand:

The primary analysis estimand for the primary endpoint with intercurrent events (ICEs) and strategies for addressing ICEs are as below. Death is not considered as an ICE. The subject with death will be regarded as non-responder.

Estimand	Analysis Population	Variable	Intercurrent Events (ICEs)	Strategy for Addressing ICEs	Measurement of interest
Primary Estimand	Full Analysis Set (FAS)	Proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days	Endovascular Thrombectomy (EVT)	Treatment Policy	All FAS participants will be included in the analysis regardless of whether they underwent EVT or not.
			Carotid endarterectomy and carotid stenting procedure	Treatment Policy	All FAS participants will be included in the analysis regardless of whether they received the procedure or not.
			Initiation of other treatment including rehabilitation or physiotherapy	Treatment Policy	All FAS participants will be included in the analysis regardless of whether they received other treatment or not.
			Discontinuation of study treatment	Treatment Policy	All FAS participants will be included in the analysis regardless of the volume of study treatment they received.
			Lost to follow-up other than death before Day 90	Hypothetical	Missing data will be considered as missing at random (MAR) and will be imputed using multiple imputation (MI) under the assumption of MAR.

Multiplicity Adjustment:

No multiplicity adjustment will be made in Part 1. In Part 2, for the preservation of the overall type I error rate at two-sided 0.05, the primary and secondary endpoints will be tested using a closed testing procedure. If the primary endpoint achieves statistical significance at two-sided 0.05, then secondary endpoints will be tested with two-sided 0.05, with their testing in a pre-specified sequential order. The detailed description of testing sequence will be provided in the Statistical Analysis Plan (SAP).

Statistical Analysis:

Unless specified otherwise, efficacy analyses will be performed on the full analysis set (FAS), which include all randomized participants who received at least one dose of study treatment (JX10 or Placebo). The primary analysis will test the null hypothesis that there is no treatment difference in the primary endpoint between participants randomized to placebo and those randomized to JX10 in the FAS. The difference in proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days between JX10 and Placebo will be analyzed using logistic regression model that include baseline covariates, namely, stroke type (M1 or proximal M2 or ICA vs. others), NIHSS assessed at the time of screening (> 10 vs. ≤ 10), and stroke onset time (4.5 to 6 hours vs. > 6 to 12 hours vs. > 12 hours).

Safety analyses will be performed on the safety analysis set which includes all participants who received at least one dose of the study drug. The number and percentage of participants reporting any treatment-emergent AEs will be coded using the MedDRA dictionary and be summarized by system organ class and preferred term.

The pharmacokinetics and pharmacodynamics analysis will be based on pharmacokinetics/pharmacodynamics dataset (PKPDS) which consists of participants who have received at least one dose of the study drug and have at least one post-baseline plasma PK and PD measurement.

Analyses will be further detailed in the Statistical Analysis Plan.

Visit Schedule:

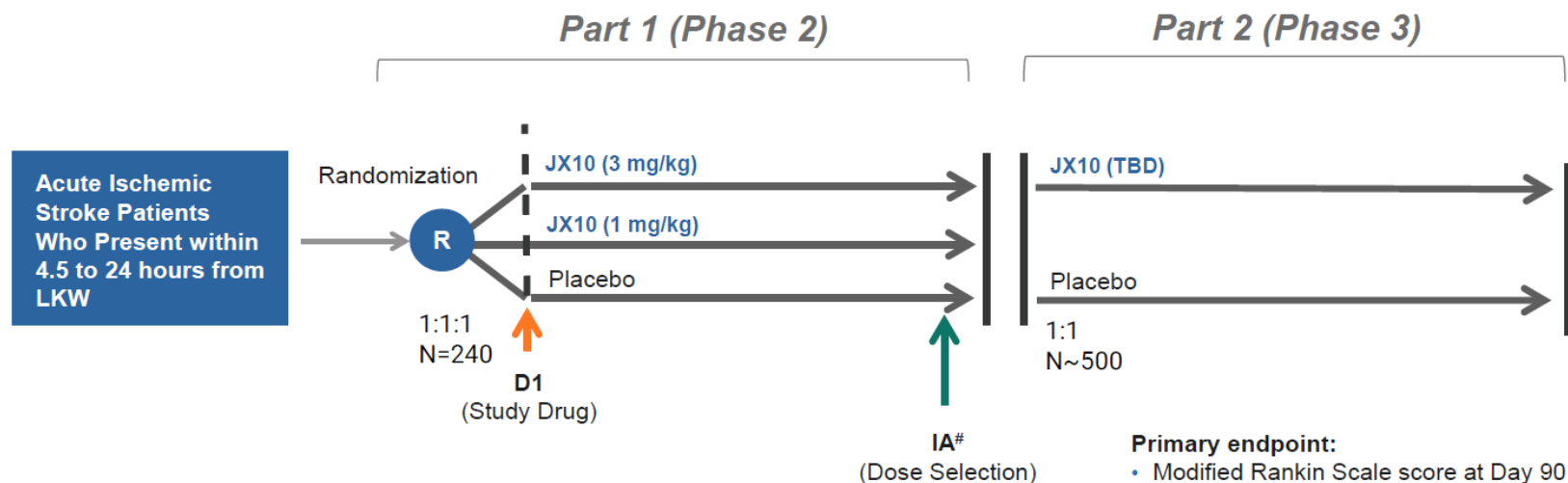
Participants will have approximately 3 visits during the study. Visit 1 includes all hospital assessments (up to and including Day 7 or day of discharge from hospital) and Visit 2 and Visit 3 consist of Day 30 and Day 90 assessments, respectively.

Study assessments conducted at each visit are listed in the Schedule of Activities ([Table 1](#)).

Duration of Study Participation:

Study duration for each participant will be approximately 90 days (or shorter in case of early participant withdrawal).

Figure 1: Schematic of Study Design



* # IA – Interim Analysis to be conducted following complete enrollment of Part 1 for dose selection to trigger start of Part 2

Investigational product will be administered as a 31-minute continuous infusion: a bolus IV dose of 10% of the dose over 1 minute, followed by 30 minutes of the remaining 90% of the dose.

Visits will occur on D7 (or at hospital discharge), D30, and D90. Telehealth or indirect assessments may be permitted in the event that participants are unable to travel to clinics.

Table 1: Schedule of Activities (for Part 1 and Part 2)

	Screening/ Enrollment	Pre-dose	Study Treatment	Post dose / End of infusion (Hours)						Day 7 or Hospital Discharge (-24hr) ^a	Day 30 (±7 days) (Telehealth /Telephone Visit)	Day 90 (±7 days) ^b	Early Term Visit ^c	Unscheduled Visit
				0 (+5 min)	1 (±5 min)	4 (±2 hr)	24 (±2 hr)	48 (±4 hr)	72 (±4 hr)					
VISITS				Visit 1							Visit 2	Visit 3		
Informed Consent	X													
Inclusion/Exclusion Criteria	X													
Demographics/Medical History	X													
Door to Study Treatment Start			X											
Time Last Known Well	X													
Concomitant Medications/Procedures ^d				X										X
Height (cm) ^e	X													
Weight (kg) ^f	X			X ^f										
Vital Signs ^g	X	X		X ^{g,h}						X			X	X
Physical Exam	X						X			X			X	X
Neurologic Exam	X						X	X	X	X			X	X
12-Lead ECG ⁱ	X			X	X		X							
Cardiac Telemetry Monitoring ^j				X										
Continuous ECG ^k		X		X										
Imaging – NCCT or MRI	X						X ^l							

	Screening/ Enrollment	Pre-dose	Study Treatment	Post dose / End of infusion (Hours)						Day 7 or Hospital Discharge (-24hr) ^a	Day 30 (±7 days) (Telehealth /Telephone Visit)	Day 90 (±7 days) ^b	Early Term Visit ^c	Unscheduled Visit
				0 (+5 min)	1 (±5 min)	4 (±2 hr)	24 (±2 hr)	48 (±4 hr)	72 (±4 hr)					
VISITS	Visit 1										Visit 2	Visit 3		
Imaging – CTA or MRA	X						X ^l							
Imaging – CTP or MRP	X						X ^l							
Imaging – mCTA ^m	X													
Pregnancy Test (Urine or Serum) ⁿ	X									X		X ^o	X ^o	
Blood Sampling for Hematology and Blood Chemistry ^p	X					X	X		X	X				
Blood Sampling for LFTs ^p	X					X	X		X	X				
Blood Sampling for Coagulation ^p	X					X	X							
Plasma Sampling for PK Analysis ^q		X ^r		X ^r	X ^r		X ^r							
Plasma Sampling for Biomarker Analysis ^q		X		X	X		X							
Urinalysis ^p		X				X	X	X	X	X				
Urine Albumin-to-Creatinine Ratio (UACR), Urine Protein-to-Creatinine Ratio (UPCR), Creatinine		X					X		X	X				

	Screening/ Enrollment	Pre-dose	Study Treatment	Post dose / End of infusion (Hours)						Day 7 or Hospital Discharge (-24hr) ^a	Day 30 (±7 days) (Telehealth /Telephone Visit)	Day 90 (±7 days) ^b	Early Term Visit ^c	Unscheduled Visit
				0 (+5 min)	1 (±5 min)	4 (±2 hr)	24 (±2 hr)	48 (±4 hr)	72 (±4 hr)					
VISITS				Visit 1							Visit 2	Visit 3		
Kinase, LDH, and Haptoglobin														
NIHSS	X ^s	X ^t		X ^u			X ^u			X			X	
Pre-stroke mRS	X ^s													
mRS	X ^s									X	X	X	X	
Barthel index score										X	X	X	X	
EQ-5D-5L												X	X	
SIS-16												X	X	
TOAST Classification ^v										X	X	X	X	
Brief Suicidal Ideation and Behavior Assessment		X								X	X	X	X	
Enrollment/ Randomization ^w	X													
Administration			X ^x											
Information collection ^y (EVT data and images), if applicable			X											
AE Recording ^z	X	X	X											X

^a The Day 7 or Hospital Discharge assessments should be conducted on Day 7 or at discharge from the hospital, whichever occurs earlier.

^b The Day 90 visit should be conducted in person whenever possible followed by telehealth or phone in order of preference. All other visits after hospital discharge can be conducted by telephone/telehealth or in person at the study participant's/representative's request.

^c Early termination is defined as withdrawal from the study. Participants who discontinue study treatment for any reason or who move to comfort care/palliative

- care should not be withdrawn from the study. Rather, they should continue protocol-required tests and assessments. Discuss with the Study Medical Monitor and/or site monitor regarding alternative methods of follow-up.
- ^d All concomitant medications and procedures should be captured from consent through end of study. Follow eCRF completion guidelines for specific instructions, especially regarding suggested methods of capturing “as needed” medications or medications with variable or intermittent dosing, or those requiring titration (e.g., heparin).
 - ^e For height assessments (taken for BMI calculations), it is sufficient to record a previously known value (for example, as recorded on a driving license) without actually measuring the participant’s height.
 - ^f Weight should be measured prior to randomization. Reliable estimated weight (based on medical record, patient/caregiver reported, etc.) will be allowed as per Investigator’s discretion prior to randomization if weight cannot be measured. If weight is measured at screening, weight does not need to be measured after randomization.
 - ^g Vital sign assessments should include blood pressure, pulse rate, oxygen saturation, respiratory rate, and temperature. Vital signs should be collected with the participant at rest and supine (or recumbent/semi-recumbent or sitting as per Investigator’s discretion). For blood pressure management refer to Section 6.4.2.
 - ^h Vital signs, including blood pressure, pulse rate, oxygen saturation, respiratory rate, and temperature, should be measured per standard of care for post-thrombolysis at least every 8 hours ($\pm$ 1 hour) for Hour 0 – 72. Following study treatment administration, measure blood pressure and perform neurologic assessment every 15 minutes ($\pm$ 5 minutes) for Hour 0 – 2, every 30 minutes ($\pm$ 10 min) for Hour 2 – 6, and hourly ($\pm$ 15 min) for Hour 6 – 24. Only abnormalities on assessment that are AEs should be recorded on the eCRF.
 - ⁱ 12-Lead ECG should be performed at screening and at 0, 1, and 24 hours post-dose. All 12-lead ECG collections (other than screening) should be as close as possible to plasma PK sampling. For participants undergoing EVT or other procedures that may temporally overlap with 12-lead ECG collection timepoints or prohibit 12-lead ECG collection, the 12-lead ECGs should be collected as close as possible to the designated time point, and a reason for delay should be entered into the eCRF.
 - ^j Cardiac telemetry monitoring should be performed for at least 24 hours following administration of study treatment.
 - ^k Continuous ECG monitoring/capture for QT evaluation to be performed in a select number of participants and sites (refer to operations manual for details).
 - ^l Where applicable and acceptable, the follow-up scan will be obtained at approximately 24 ± 6 hours after study drug administration. MRA/MRP is preferred but CTA/CTP would be acceptable. The MR scan (if available and feasible) should include an MRA of the Circle of Willis to assess for recanalization and MRP to assess for reperfusion in addition to standard fluid-attenuated inversion recovery, gradient recalled echo, and DWI sequences per standard of care.
 - ^m mCTA may be performed as an alternative to CTP for imaging-based eligibility.
 - ⁿ Pregnancy testing (urine or serum) is to be performed only in women of childbearing potential. Participants with a negative pregnancy test (urine or serum) result will be randomized in the study.
 - ^o Serum pregnancy test from local laboratory is only required in women of childbearing potential if a delay in menstrual period is confirmed during telehealth, phone, or in-person visit.
 - ^p Blood sampling for hematology, blood chemistry, LFTs, coagulation tests, pregnancy tests, and urine sampling for urinalysis will be analyzed at the local laboratory. Finger stick blood glucose is acceptable for screening purposes. Urinalysis is applicable to participants who can provide a urine sample spontaneously or, if planned, via catheter placement.
 - ^q Plasma sampling for PK and Biomarker assessments should be collected and sent to the central laboratory.
 - ^r There are total of 4 PK samples at pre-dose, 0 hour post-dose/end of infusion, 1 hour post-dose/end of infusion, and 24 hours post-dose/end of infusion. Plasma sampling for the 0- and 1-hour post-dose PK assessments should be done as close to these timepoints as possible. For details refer to Section 8.1.3.
 - ^s Study-specific qualified rater will not be required for screening.

- ^t NIHSS score should be reassessed pre-dose (as close to before the start of study treatment as possible) for those participants who have an elapsed time of ≥ 60 minutes from the time the NIHSS was assessed at screening up to the anticipated time of the start of the study treatment.
- ^u The post-dose NIHSS assessment should be performed as soon as possible after completion of the study treatment infusion but may be performed up to 24 hours after completion of study treatment infusion.
- ^v TOAST classification may be performed anytime up to Day 7 or hospital discharge (whichever occurs earlier), Day 30, and Day 90.
- ^w Participants will be randomized/enrolled after all screening assessments have been completed and after the Investigator has verified all eligibility criteria have been evaluated and documented in the participant's source that the participant is eligible to participate.
- ^x Administration of study treatment should be initiated as soon as possible following receipt of prepared study treatment and within 6 hours of last known well or within 24 hours of last known well with qualifying penumbral volume on perfusion or eligible imaging modality. *For participants undergoing EVT, allow for an hour between start of study drug infusion and start of endovascular procedure (e.g., femoral puncture).*
- ^y For EVT data and imaging collection, see Section 4.1.1.
- ^z At each study visit, the Investigator will assess the participant for AEs and will record any new AEs or updates to previously reported AEs. At any point in the study, unscheduled imaging and other assessments can and should be collected especially if they are performed due to AEs.

2. INTRODUCTION

JX10 is under development for the treatment of acute ischemic stroke (AIS) and may address a critical unmet medical need for those AIS patients whose symptom onset is ≥ 4.5 hours.

2.1. Overview of AIS

Stroke is a major global public health problem and is the second leading cause of death worldwide. Ischemic stroke cases account for 62% to 87% of total stroke cases [GBD 2019 Stroke Collaborators 2021; Tsao 2022]. Globally, there were 7.8 million incident ischemic stroke cases, 69.9 million prevalent ischemic stroke cases, 3.7 million deaths, and 70.4 million disability-adjusted life-years due to ischemic stroke in 2019 [Martin 2024; Li 2024]. In the US, there are approximately 700,000 new or recurrent ischemic stroke cases each year (calculated based on the percentage of ischemic stroke cases [87%] that contribute to all stroke cases [795,000] [Martin 2024]. A major challenge with existing thrombolytic therapies approved for AIS is their limited time window for treatment. Delay in hospital presentation results in limited access to the treatment. Most of the patients present late which leads to less availability and effectiveness of reperfusion therapies and worse outcomes after stroke [Dhand 2024].

2.1.1. Pathophysiological Mechanism of Ischemic Stroke

Ischemic strokes are caused by sudden interruption or significant impairment of blood and oxygen supply to the brain which results in corresponding loss of neurological function. TOAST classification is used to categorize subtypes of AIS based on etiology: large artery atherosclerosis, cardioembolism, small-vessel occlusion, stroke of other determined etiology, and stroke of undetermined etiology [Adams 1993].

- Large-artery atherosclerosis category of AIS is generally caused by occlusion of a major brain artery or a branch cortical artery.
- Cardioembolic strokes are observed in patients with arterial occlusions that are embolic in source, often from atrial fibrillation.
- Small-vessel occlusion (lacune) AIS are mostly seen in patients with history of chronic conditions like diabetes, hypertension, hyperlipidemia, or lipohyalinosis, which can lead to occlusion of the lumen of smaller vessels.
- AIS of other determined etiology is generally caused by inherited diseases, metabolic disorders, and coagulopathies.
- AIS of undetermined etiology is confirmed by imaging findings that suggest embolism, but etiology cannot be determined and excludes all the above categories.

2.1.2. Risk Factors for Stroke

The risk of AIS is influenced by both modifiable and non-modifiable risk factors. Common non-modifiable risk factors include age, family history of stroke, race, gender, and history of prior ischemic stroke. Modifiable factors include hypertension, dyslipidemia, smoking, and physical inactivity. Primary and secondary stroke prevention strategies aim to reduce the risk of ischemic stroke by targeting modifiable risk factors [Kleindorfer 2021; Meschia 2014].

2.1.3. Current Therapies for AIS

The early management of AIS focuses on timely reperfusion, prevention of future strokes, and supportive care. Reperfusion therapies include thrombolysis using rtPA (e.g., alteplase) and EVT with mechanical thrombectomy. Use of available reperfusion therapies is limited by the short time window for treatment and by the selective candidacy criteria applied to patients. Supportive care consists of hemodynamic support, glycemic control, nutritional support, temperature management, and typically admission to a dedicated hospital stroke unit with specialized nursing care.

Thrombolysis with Activase (alteplase) is the first-line reperfusion therapy for AIS, but its use is limited to within a 3- to 4.5-hour window after AIS onset because of limited efficacy with excess risk of hemorrhagic complications, including symptomatic intracranial hemorrhage. At select centers, with eligible perfusion imaging, alteplase has been administered up to 9 hours from Last Known Well (LKW) although the benefits are uncertain [Hollist 2021].

Alteplase is approved for use in AIS within 3 hours of symptom onset in most countries and regions, including the US (1996), EU (2002), and Japan (2005). The use of alteplase for AIS is supported by international guidelines [Berge 2021; Powers 2019]. The most frequent adverse reaction to alteplase is bleeding. Additional adverse reactions of alteplase include hypersensitivity, thromboembolism, and cholesterol embolization [ACTIVASE USPI 2018]. In the US post-marketing experience with alteplase for the AIS indication, the following adverse reactions have been identified: cerebral edema, cerebral herniation, seizure, new ischemic stroke, and embolism [ACTIVASE USPI 2018]. Notably, the window in which alteplase is recommended encompasses a limited proportion of AIS patients, and an even smaller group are tPA-eligible because of relative and absolute contraindications to administration or rapid improvement of symptoms [Mac Grory 2021].

Recently, tenecteplase has been approved in the US by the US FDA for treatment of acute stroke within the first 3 hours of presentation based on the results from AcT trial conducted in Canada [Menon 2022]. In the TIMELESS trial, tenecteplase failed to best placebo among the AIS patients with salvageable tissue (n = 458), as determined by perfusion imaging, who received study drug between 4.5 and 24 hours of LKW (adjusted common odds ratio for the distribution of scores on the modified Rankin scale at 90 days 1.13, tenecteplase vs. placebo (95% confidence interval, 0.82 to 1.57; P=0.45). A lack of clinical benefit was observed in spite of a nominally higher rate of recanalization in the tenecteplase arm (76.7%, vs. 63.9% in the placebo

arm). No safety advantage was observed with tenecteplase, with symptomatic ICH (sICH) occurring in 3.2% vs 2.3% of patients, tenecteplase vs. placebo, respectively [Albers 2024].

Mechanical thrombectomy (MT) is an effective alternative to medical treatment, especially in large vessel occlusions (LVO) that make up 15% to 30% of acute ischemic strokes [Rai 2017; Waqas 2020]. This is supported by findings from studies such as DAWN and DIFFUSE-3 that have shown benefits in use of MT in AIS patients presenting up to 24 hours, with large vessel occlusion and perfusion imaging evidence of mismatch [Nogueira 2018; Albers 2018], and confirmed recently in the SELECT 2 trial [Sarraj 2023]. International guidelines support this approach; in particular American Heart Association/American Stroke Association guideline states that “in selected patients with AIS within 6 to 16 hours of last known normal who have LVO in the anterior circulation and meet other DAWN or DEFUSE 3 eligibility criteria, MT is recommended. In selected patients with AIS within 16 to 24 hours of last known normal who have LVO in the anterior circulation and meet other DAWN eligibility criteria, MT is reasonable [Powers 2019]. The European Stroke Organization also recommends that “in adults with anterior circulation large vessel occlusion-related acute ischemic stroke presenting between 6 and 24 hours from time last known well and fulfilling the selection criteria of DEFUSE-3 or DAWN, mechanical thrombectomy plus best medical management over best medical management alone to improve functional outcome” [Turc 2023]. Efforts to extend MT to medium vessel occlusions; however, have failed with recent studies (DISTAL, DISCOUNT, ESCAPE-MeVO) not only failing to demonstrate benefits, but saw potential harm [Clarencon 2024; Goyal 2025; Psychogios 2025]. Thus, despite the potential efficacy of MT, there is urgent need for better treatment for the majority of patients who lack access or are not candidates for thrombectomy.

There is clearly an urgent unmet need for the development of an effective thrombolytic drug that can be administered in an extended window, without increasing hemorrhagic liability, to provide a therapeutic option for the medical management of AIS in patients with little options.

2.2. Profile of Previous Experience with JX10

2.2.1. Nonclinical Experience

In nonclinical studies, the pharmacological activity of JX10 was evaluated in animal models of cerebral infarction, including rodent, gerbil, and monkey models. Reduction of infarct volume as well as suppression of hemorrhagic conversion were observed across all models, with improved hemorrhage rates relative to alteplase. In both the thrombotic and embolic cerebral infarction models in monkeys, administration of JX10 led to a reduction in infarct volume. In the thrombotic cerebral infarction model, neurological deficits were improved, and hemorrhagic transformation was suppressed. In the embolic cerebral infarction model, JX10 administration showed superior reduction of infarct volume, compared with alteplase. Overall, these findings show that JX10 is an effective thrombolytic drug, which, based on preclinical data, may be effective in the thrombolytic treatment of AIS. In addition, this drug has notable anti-inflammatory and antioxidant activity and minimal effect on bleeding time/hemostasis which could be advantageous when compared to other available therapies [Sawada 2014].

2.2.2. Clinical Experience

JX10 clinical development to date has consisted of a Phase 1 study (TMS-007-1) in healthy Japanese adult male participants aged 20 to 45 years and a Phase 2a study (TMS-007-2) in Japanese participants with AIS within 12 hours of onset. All clinical studies were conducted in Japan under a Clinical Trial Notification with the PMDA and complied with GCP.

TMS-007-1 The Phase 1 study (TMS-007-1) was a single-dose study to assess the safety and tolerability of JX10 in healthy Japanese male volunteers. A total of 30 participants received JX10 (3, 15, 60, 180, or 360 mg) in this study and 10 participants received placebo. The study treatment (JX10 or placebo) was administered intravenously (10% bolus for 1 minute, followed by 90% IV infusion over 30 minutes).

At 30 minutes post-infusion, the plasma concentration of JX10 had decreased to less than a quarter of the peak level and gradually declined thereafter. T_{max} occurred at the end of infusion ranging from 31.0 to 31.17 minutes. The mean $t_{1/2}$ ranged from 12.9 to 15.4 minutes with no significant difference between cohorts. Both C_{max} and AUC_{0-24} after JX10 administration increased in a dose-proportional manner. The estimated slope (95% CI on both sides) using the power model was 1.03 (0.99-1.07) for AUC_{0-24} and 1.02 (0.98-1.05) for C_{max} , both of which showed dose proportionality.

The incidence of adverse events (AEs) was generally similar in all cohorts. All AEs were mild in severity, not serious, and all AEs recovered by the end of the study, except for 1 AE (injection site phlebitis) that was assessed as recovering. Moreover, there were no deaths or discontinuations of treatment due to AEs.

Injection site phlebitis and hemorrhage are potential risks of JX10. Injection site phlebitis was observed in the Phase 1 study (TMS-007-1) but not in the Phase 2a study (TMS-007-2). In Study TMS-007-1, the AEs of injection site phlebitis were mild in severity and occurred in the highest dose cohorts.

TMS-007-2 The Phase 2a (TMS-007-2) single-dose study evaluated the safety and efficacy of JX10 administered to participants with cerebral infarction within 12 hours of symptom onset. A total of 52 participants received JX10 (6 received 1 mg/kg JX10, 18 received 3 mg/kg JX10, and 28 received 6 mg/kg JX10) and 38 participants received placebo. The maximum dose in each cohort was capped based on a body weight of 60 kg. The study drug (JX10 or placebo) was administered to each participant intravenously (10% bolus for 1 minute, followed by 90% IV infusion over 30 minutes).

Average time to treatment in this ascending-dose, double blind, cohort study of doses up to 6 mg/kg (maximum dose 360 mg) was 9.5 hours in the JX10 group (n=52) vs. 9.3 hours in the placebo group (n=38). Overall revascularization rates of angiographic occlusions were 58.3% in participants treated with JX10 vs. 26.7% for placebo-treated participants, based on an improvement relative to baseline of 24-hour Arterial Occlusive Lesion (AOL) scores. While baseline Alberta Stroke Program Early CT Score (ASPECTS) and National Institutes of Health

Stroke Scale (NIHSS) scores were balanced at baseline, rates of patients with no or minimal symptoms, based on a 90-day mRS score of 0-1, were 40.4% vs. 18.4% for JX10 versus placebo, respectively. No sICH was observed in the JX10 treatment arms. See [Figure 2](#) for distribution of mRS between JX10 and placebo and [Figure 3](#) for summary of key results.

The incidence of AEs, SAEs, and AEs leading to discontinuation was generally similar between the JX10-treated groups and the placebo group. In the clinical program, 3 deaths were reported, all in Study TMS-007-2, including 1 JX10-treated participant and 2 placebo-treated participants. All SAEs resulting in death were assessed as not related to JX10.

These data demonstrated a favorable safety profile, high rates of revascularization of intracranial occlusions assessed via MRA, and a statistically significant improvement of the proportion of participants with functional independence at 90 days compared with placebo.

Figure 2: mRS outcome on Day 90

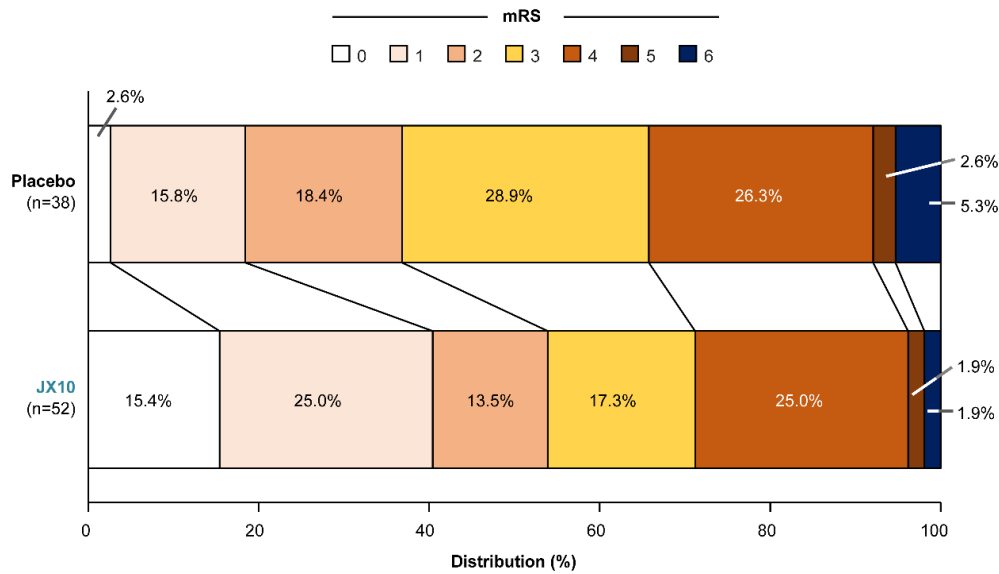
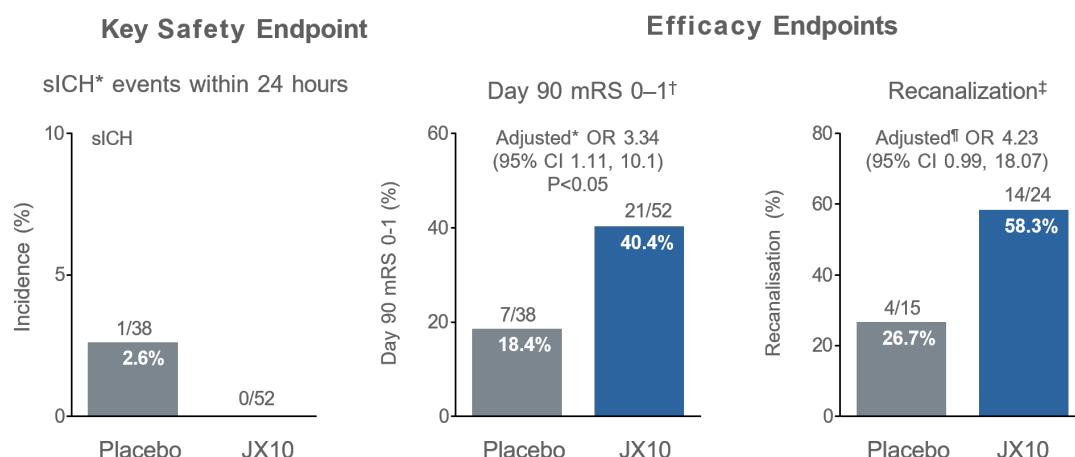


Figure 3: Key Results



*Defined as type 1 and 2 parenchymal hemorrhage with a deterioration of ≥ 4 points in the NIHSS through 24 hours after administration; adjusted OR for binary response of mRS 0–1 vs 2–6; †recanalization of AOL score of 2–3 for the subset of patients who had visible occlusion on MRA at baseline; ‡calculated from the logistic regression with covariates (treatment and participant subgroup) adjustment; JX10 refers to pooled data from all dose groups. AOL, arterial occlusive lesion; CI, confidence interval; ICH, intracerebral hemorrhage; MRA, magnetic resonance angiography; mRS, modified Rankin Scale; OR, odds ratio; sICH, symptomatic ICH

See the Investigator’s Brochure for detailed information on relevant nonclinical and clinical studies.

2.3. Study Rationale

As discussed above, Stroke is a major global public health problem and is the second leading cause of death worldwide [GBD 2019 Stroke Collaborators 2021]. In the US alone, there are approximately 700,000 new or recurrent ischemic stroke cases each year (calculated based on the percentage of ischemic stroke cases [87%] that contribute to all stroke cases [795,000]) [Martin 2024]. However, current standard of care is limited in terms of options and treatment window. This problem is evident in the results from the US GWTG-Stroke registry which showed that among the 39 760 patients with AIS less than 24 hours after LKW, 51.2% arrived after the alteplase window and 48.8% arrived within the alteplase window but were not treated. [Mac Grory 2021]. Substantial number of patients worldwide are left with little treatment options. There is a need for development of thrombolytic drugs with a longer treatment window and an improved safety profile that would address this serious unmet medical need.

JX10 is a thrombolytic agent with a novel mechanism of action of inducing conformational changes in plasminogen to increase downstream fibrin affinity, as opposed to the systemic plasminogen activation characteristic of alteplase and tenecteplase. The resultant activity is thrombosis-specific physiological fibrinolysis. In addition to this effect, JX10 may have favorable anti-inflammatory activity through its inhibition action on soluble epoxide hydrolase (sEH) and antioxidant activity based on its chemical structure, which may suppress hemorrhagic changes associated with cerebral infarction [Sawada 2014]. The anti-inflammatory effects of

JX10 are thought to reduce the risk of intracranial hemorrhage associated with treatment, via a reduction of endothelial cell damage at the site of thrombosis and preservation of vascular integrity [Hasumi 2021]. Animal models and clinical data to date support this hypothesis. This combined mechanism of action of JX10 is thought to induce fibrinolysis that helps restore cerebrovascular circulation, while preventing and mitigating potential hemorrhage conversions associated with cerebral infarctions. Thus, JX10 has the potential to overcome current challenges associated with existing thrombolytic agents by offering effective fibrinolysis with an extended treatment window while minimizing hemorrhage risks.

JX10 has encouraging early clinical data that suggest potential benefits in patients who presented late following AIS, up to 12 hours from LKW. In the Japanese Phase 2a Study (TMS-007-2), JX10 compared to placebo among Japanese AIS patients presenting within 4.5 and 12 hours from LKW, JX10 had higher rates of recanalization and significantly more patients with better neurologic outcomes, as evidenced by the proportions of participants with no or minimal symptoms (modified Rankin Scale (mRS), 0-1) by 90 days.

The design of the current study also takes into account lessons and insights from the recently completed TIMELESS trial in a similar population [Albers 2024] and findings from that showed that the volume of penumbral tissue in large vessel occlusion diminishes significantly over time, especially by 15 hours from LKW [Seners 2023], to ensure proper selection of study population.

Thus, this global study will aim to evaluate the use of this novel thrombolytic JX10 in a broader population of participants with AIS who present between 4.5 to 24 hours [(1) between 4.5 and 6 hours after LKW, if no perfusion imaging is available, or (2) between 4.5 and 24 hours from LKW, if perfusion or equivalent imaging shows salvageable penumbra].

2.4. Rationale for Dosing Regimen

Study TMS-007-1 was the first clinical trial conducted in healthy human volunteers. Study TMS-007-2 was the second clinical trial, a Phase 2a study conducted in Japan which demonstrated efficacy (revascularization and benefits on 90-day mRS scores) with JX10 compared with placebo with pooled data from all dose groups. All three doses (1, 3, and 6 mg/kg) showed trends of improvement over placebo, with the 3 and 6 mg/kg cohorts having fairly comparable nominal efficacies based on rates of recanalization and improvements in neurological outcomes, as measured by mRS. However, given the small sample size, the relationship between dose or exposure and efficacy measures could not be established.

The doses for the current study are described below:

- 3 mg/kg (maximum dose of 300 mg) a dose that showed a trend of efficacy, comparable to what was observed with the 6 mg/kg dose in TMS-007-2.
- 1 mg/kg (maximum dose of 100 mg).

2.5. Benefit-Risk Assessment

The potential risks related to participation in this study are justified by the anticipated benefit to participants.

Detailed information about the known and expected benefits and risks and reasonably expected AEs of JX10 is provided in the Investigator's Brochure and informed consent form. A high-level summary of those benefits and risks known during study design is provided here.

AIS is associated with disability and mortality. Treatment options for AIS beyond 4.5-hour time window are limited both in availability and applicability to select stroke centers and select patients.

JX10 has the potential to provide patients and physicians with a new treatment option that may reduce functional disability associated with AIS. In nonclinical models, JX10 demonstrated thrombolytic activity by promoting recanalization, reducing infarct size, and improving neurologic function.

Early clinical development of JX10 demonstrated efficacy in the revascularization of both medium and large vessels when assessed within 24 hours following treatment. Treatment with JX10 in Study TMS-007-2 was also associated with functional improvement based on the mRS. These data represent a potential novel treatment option for patients with occlusions that are not amenable to endovascular thrombectomy presenting outside the currently approved thrombolytic windows.

The JX10 formulation includes the excipient polyoxyethylene hydrogenated castor oil 60 (HCO-60)/polyoxyl hydrogenated castor oil/macrogolglycerol hydroxystearate, which has a known risk of anaphylaxis.

In this study, as JX10 is evaluated in combination with standard of care vs. standard of care, placebo is being used to maintain the blind. The use of placebo is justified as there are no approved or confirmed pharmaceutical treatment options for AIS beyond the window of 4.5 hours from symptom onset.

The start of the treatment window (i.e., from 4.5 hours from LKW) for the study may vary in compliance with local or regional guidelines for thrombolytic treatment. All participants will receive standard of care treatment based on applicable local guidelines.

The potential benefits of participation in this study are expected to outweigh the risks.

3. STUDY OBJECTIVES AND ENDPOINTS

The objectives and endpoints for Part 1 and 2 of this study are listed in [Table 2](#).

Table 2: Objectives and Endpoints

Primary Efficacy Objective	Primary Efficacy Endpoints
To determine if JX10 improves functional outcome as measured by the mRS when compared with placebo following AIS	Proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days
Secondary Efficacy Objective	Secondary Efficacy Endpoints
To evaluate the effects of JX10 on acute and 90-day clinical outcomes	Ordinal mRS score, based on a 6-point ordinal scale at 90 days Proportion of participants with functional independence (mRS score 0-2) at 90 days Proportion of participants with BIS > 90 at 90 days Proportion of participants with favorable outcome (mRS 0-1 or 0-2), taking into account baseline NIHSS score (mRS sliding dichotomy) Proportion of participants with reperfusion at 24 hours post-treatment, defined as > 90% reduction in $T_{max} > 6$ seconds lesion volume or complete recanalization on CT angiography (CTA)/magnetic resonance angiography (MRA) Proportion of participants with recanalization at 24 hours post-treatment, defined as complete or partial recanalization on CTA/MRA Proportion of participants with angiographic reperfusion (TICI 2b/3 or TICI 3) at completion of angiographic procedure (endovascular thrombectomy patients only) Proportion of participants where planned thrombectomy was not performed (planned endovascular thrombectomy patients only) Proportion of participants with improvement on the NIHSS by > 7 points or score 0 or 1 at 7 days or hospital discharge

Primary Safety Objective	Primary Safety Endpoints
To evaluate the risk of symptomatic intracranial hemorrhage of JX10 in participants with acute ischemic stroke	Incidence of symptomatic intracranial hemorrhage, defined as local or remote parenchymal hemorrhage type 2, subarachnoid hemorrhage, and/or intraventricular hemorrhage within 36 hours post-randomization, combined with a neurological deterioration of 4 points or more on the NIHSS from baseline (the closest collection before administration of the study treatment), or from the lowest NIHSS value between baseline and 24 hours, or leading to death
Secondary Safety Objective	Secondary Safety Endpoints
To evaluate the safety and tolerability of JX10 in participants with acute ischemic stroke	Incidence of AEs and SAEs. Incidence of any intracranial hemorrhage within 24 hours of study treatment Incidence of any intracranial hemorrhage within 14 days of study treatment Incidence of major bleeding within 24 hours of study treatment Incidence of major bleeding within 14 days of study treatment Incidence of symptomatic cerebral edema within 14 days of study treatment Proportion of participants with parenchymal hematoma type 2 within 7 days following treatment based on Heidelberg bleeding classification Proportion of participants with parenchymal hematoma type 1 or 2 within 7 days following treatment based on Heidelberg bleeding classification Overall mortality rates at 30 and 90 days
Tertiary Objectives	Tertiary Endpoints
Imaging Objective	
To evaluate the effects of JX10 on arterial revascularization (among participants in evaluable paired imaging studies)	Proportion of participants with arterial revascularization or reperfusion of the ischemic field: <i>Revascularization of occluded intracranial arteries is defined by an AOL score of 2 or 3 on CTA or MRA after treatment completion</i>

	OR <i>For participants with no visible intracranial occlusion on CT angiography at baseline, > 90% reversal of the baseline T_{max} > 6 seconds lesion after treatment completion</i>
To evaluate the effects of JX10 on final infarct volume	Proportion of penumbral tissue salvaged (non-progression to infarction) at 24 hours Final infarct volume at 24 hours by MRI or NCCT, corrected for baseline ASPECTS
Quality of Life	
Quality of life and health resources measures	Stroke Impact Scale -16 Questions at 90 days EuroQOL-5-level assessment of health outcomes at 90 days Health Resources Utilizations at 90 days
Biomarkers	
Plasma biomarkers supportive of target engagement	Change from baseline in DHET, EET, plasmin antiplasmin complex, alpha 2 antiplasmin
Pharmacokinetics	
Pharmacokinetic profile of JX10	JX10 concentration in plasma (exposure-response evaluations)

4. STUDY DESIGN

4.1. Study Overview

This is a randomized, multicenter, double-blind, placebo-controlled study designed to maximize operational efficiency modeled after established studies like the NINDS trial to dose-find and confirm the efficacy and safety of that dose in an appropriately sized study to support eventual regulatory registration of JX10. The study is intended to test whether a novel thrombolytic with anti-inflammatory actions would benefit a broad group of patients with AIS who present for care, within 4.5 to 24 hours. The enrollment and participation in the trial will not preclude or delay standard of care treatment including EVT.

Approximately 150+ sites are planned globally.

The study will consist of 2 parts:

Part 1 will be a dose-ranging study to identify the optimal dose of JX10 based on assessment of safety and efficacy. A subset of participants with qualified imaging studies will be evaluated for angiographic revascularization. To establish the optimal dose in Part 1 of the study, participants will be randomized 1:1:1 to receive JX10 at dosing levels of 1 mg/kg, 3 mg/kg, or placebo, with approximately 80 participants in each arm. The primary efficacy outcome measure will be mRS at Day 90.

An interim analysis will be performed when the last participant in Part 1 has been enrolled. Based on best available data at the time of the interim analysis, a determination will be made by the independent data monitoring committee (IDMC) as to whether there is a dose that will be advanced to Part 2. The optimal dose of JX10 for Part 2 will be selected based on the totality of data, including radiographic, clinical, safety, and PK data from Part 1. The most efficacious dose with an acceptable safety profile will be selected as the dose to advance to Part 2. The acceptance of the recommendation will trigger the start of enrollment for Part 2, with site notification. Should no dosing arm demonstrate superiority relative to the placebo group, the Sponsor may decide to delay the start of Part 2 or terminate the study.

Part 1, upon the last participant completing the last visit, will close out, unblind, and report results when the last participant has completed the final visit. Regulatory interactions will proceed as needed. Select centers may have delayed start of enrollment into Part 2, pending additional approval or authorization by responsible regulatory authorities. Part 2 may be amended based on findings of Part 1 and requirements from respective regulatory authorities. Part 1 participants will not contribute to the analysis of Part 2. However, pertinent findings and learnings from Part 1 will be used to inform and potentially modify Part 2, including but not limited to modifications of endpoints, procedures, eligibility criteria, and sample size.

In Part 2 of the study, approximately 500 participants may be randomized 1:1 to receive either JX10 (optimal dose selected after the end of Part 1 analysis) or placebo. Enrollment of Part 2 will start, where permitted, while Part 1 completes the follow-up and database lock. The primary

efficacy measure for Part 2 of the study is the same mRS outcome at Day 90. The primary analysis for Part 2 will include only participants enrolled in Part 2 of the study.

Participants who meet all eligibility criteria and provide a signed informed consent will be randomized to the JX10 or placebo arms. Participants are to receive the study drug administration as soon as possible with the prespecified treatment window, within 6 hours from LKW among participants with no perfusion imaging or within 24 hours with qualifying penumbral volume on perfusion or eligible imaging modality.

The study is designed to enroll a broad population across various hospital types, regions, and ischemic stroke subtypes.

- To minimize bias, enrollment of participants with very late presentations (16 to 24 hours from LKW) may be limited.
- Planned thrombectomy is permitted as per guidelines in the ICA, M1, and proximal M2, if it is performed at least one hour or more, from the beginning of the study drug administration. Sites should only enroll participants who 1) do not have ready access to EVT, 2) are ineligible to EVT, or 3) due to a lack of immediate availability of EVT (such as need for hospital transfer, etc.), are anticipated to have at least an hour between the beginning of study drug administration and start of EVT. The number of participants who receive study drug administration followed by planned EVT should constitute no more than 30% of all randomized participants. As data accrue, the Sponsor, in consultation with IDMC, may adjust the enrollment limits. Unplanned rescue procedures are not subject to restrictions.

Participants will be monitored closely, especially in the first 24 hours, during which frequent blood pressure and neurological assessments will occur. Usual precautions should be exercised in participants who will undergo possible thrombectomy at least 1 hour following the initiation of study drug administration. Participants are followed for at least 90 days. Outcome assessments such as mRS or NIHSS should be performed by trained and certified professionals. Every effort should be made to complete all protocol-mandated procedures and evaluations. In recognition of the potential need for extended hospital stay, rehabilitation, mobility, and transportation issues, telehealth options are permitted.

See [Figure 1](#) for a schematic of the study design.

4.1.1. Neuroimaging Timing and Procedures

At baseline to determine trial eligibility, comprehensive or primary stroke care imaging centers will perform qualifying imaging studies of CT/MRI, CTA ± CTP or MRA ± MRP. Centers unequipped to perform CTP, or with a preference for mCTA for enrollment, will be permitted to perform mCTA to assess collateral scoring instead of target mismatch. Eligibility on CTP will be

based on a target mismatch profile of ratio > 1.2 , core volume < 70 mL, and mismatch volume > 10 mL.

Where possible, study participants will then have final infarct volume/ICH assessment at 24 ± 6 hours, following treatment by either NCCT or MRI. Follow-up scans with MRA + MRP are preferred. Participants with paired and evaluable studies will be evaluated for rates of successful recanalization.

Digital images including but not limited to digital subtraction angiography (DSA) images from diagnostic angiography (baseline, mechanical thrombectomy, and post thrombectomy images obtained during angiogram for assessment of TICI scores) will be submitted centrally for independent blinded analysis.

4.1.2. Assessments and Evaluation of Neuroimaging

Briefly, baseline assessments for all participants enrolled will include ICH, ASPECTS, target mismatch and core volumes, site of occlusion (if available), and, if applicable, pial vessel extent and/or grading.

At 24 hours, when possible, participants will undergo MRI (T2*, FLAIR, DWI) or NCCT to assess final infarct volume, and calculation of penumbral tissue salvage.

Recanalization will be assessed centrally using the AOL score.

To ensure efficacy read quality, the CT or MRI data collected at the recruiting centers, will be electronically transferred, and read at a central imaging center on a pre-determined basis.

Individual centers will be qualified at the start and ongoing through the trial and will be responsible at the time of participant screening for subject enrollment.

4.2. Study Duration for Participants

The total study duration for each participant will be approximately 90 days (and shorter in case of early study termination):

- Treatment period of 31 minutes (after a brief Screening – to allow for study treatment initiation within 24 hours of the LKW).
- Follow-up period of 90 days.

Participants will have approximately 3 visits during the study. All visits should be performed within the time window specified in the Schedule of Assessments. Visit days are calculated with respect to Day 1 (the date of the end of study treatment administration).

The end of study date for a participant may be the last study visit, last follow-up telephone conversation, or last protocol-specified assessment, or if the participant has ongoing SAEs that are being followed, the date of SAE resolution.

4.3. Study Stopping Rules

The Sponsor may terminate this study at any time, after informing Investigators. The Sponsor will notify Investigators when the study is to be placed on hold, completed, or terminated. Dosing may be terminated by the Sponsor at the recommendation of the IDMC based exclusively on safety and tolerability data. In addition, the Sponsor may terminate the study at any time due but not limited to the following reasons:

- If benefit-risk assessment becomes unacceptable owing to, for example.
 - New or emerging safety findings from this study
 - Results of any interim analysis,
 - Results of review by the IDMC
- If the study conduct (e.g., recruitment rate, drop-out rate, data quality, protocol compliance, etc.) does not suggest a proper completion of the study within a reasonable time frame
- If there is a significant product issue (e.g. withdrawal of the license to the manufacturer or of the permission to import, etc.)

4.4. Unscheduled Visits

Data collected during unscheduled visits should be recorded on CRFs only for assessments indicated in [Table 1](#).

4.5. End of Study

The end of Part 1 is the last participant's Day 90 visit during Part 1. The end of study is the last participant's Day 90 visit during Part 2.

5. STUDY POPULATION

The study is intended to evaluate a new therapeutic option for those patients with AIS who present late for care, beyond the 4.5 hour recommended treatment window of conventional thrombolytic treatment, i.e., alteplase. To be eligible to participate in this study, candidates must meet the following eligibility criteria at Screening/Enrollment, or at the timepoint specified in the individual eligibility criterion listed.

5.1. Inclusion Criteria

1. Age ≥ 18 and ≤ 90 years old (Age > 85 must be mRS 0 at baseline (premorbid) to be eligible).
2. Written informed consent by patient, his/her legally authorized representative, or independent physician where local regulation allows (in accordance with all local and national regulations or according to the local ethics committee's guidelines or by another process compliant with applicable national laws and regulations and ethics committee requirements).
3. Acute ischemic stroke with compatible clinical presentation and symptomatic high grade or complete occlusion of the intracranial internal carotid, M1, M2, or distal branches of the middle cerebral artery (MCA), anterior cerebral artery (ACA), or posterior cerebral artery (PCA), demonstrated by:
 - a. Computer Tomography Angiography (CTA) or Magnetic Resonance Angiography (MRA) with high grade stenosis or occlusion of an eligible artery, or
 - b. Computer Tomography or Magnetic Resonance Perfusion (CTP/MRP) with focal deficit in an eligible artery, or
 - c. Non-contrast Computer Tomography (NCCT) with hyperdense artery sign in an eligible artery or Magnetic Resonance Imaging (MRI) with susceptibility vessel sign (defined as a hypointense signal exceeding the diameter of the contralateral artery at the thrombus site) in an eligible artery.
4. Radiographic evidence of salvageable tissue* is present based on:
 - a. A mismatch ratio (perfusion lesion ($T_{\max} > 6$ seconds) volume/core estimate) > 1.2 ,
 - b. Estimated core infarct volume < 70 mL, and
 - c. Total mismatch volume ≥ 10 mL on CTP or magnetic resonance (MR) perfusion-weighted imaging (PWI)/MRI diffusion imaging.

The perfusion criteria apply to all participants, including those who present within 4.5 to 6 hours of LKW, if perfusion imaging is obtained. Perfusion is mandatory for patients presenting beyond 6 hours, optional for patients < 6 hours.

**The quantitative assessment of penumbra may include alternative methods (e.g., if the software uses different parameters in place of $T_{\max}$).*

5. Presentation with intended study treatment within 4.5 to 6 hours of Last Known Well (LKW) in the absence of radiographic assessment of perfusion or within 4.5 to 24 hours of LKW with radiographic evidence of penumbra meeting criterion (4).
6. Pre-treatment score of NIHSS ≥ 5 .
7. Functionally independent prior to stroke onset as evidenced by premorbid mRS < 2 (< 1 if age 86-90).
8. All women of childbearing potential and all men must practice contraception as described in Section 8.3.5. For women of childbearing potential, a negative pregnancy test at screening is required. All women of childbearing potential must practice effective contraception for at least 30 days after their last dose of study treatment. All men must practice effective contraception during the study and for 90 days after their last dose of study treatment. In addition, participants should not donate sperm or eggs during the study and for at least 90 days after their dose of study treatment.

5.2. Exclusion Criteria

Neurovascular Imaging

1. Radiographic findings pre-randomization of any of the following:
 - a. Large core infarction, evidenced by a core infarct volume > 70 mL, assessed on DWI or CTP; or extensive early ischemic change (hypodensity) on non-contrast CT estimated to be $> 1/3$ MCA territory, or significant hypodensity outside the $T_{\max} > 6$ seconds perfusion lesion that invalidates mismatch criteria, or
 - b. Occlusion in more than 1 vascular territory confirmed on CTA/MRA, or
 - c. Significant mass effect or clinically significant cerebral edema per Investigator's judgement, or
 - d. Evidence of acute intracranial or extracranial hemorrhage, intracranial tumor (except small meningioma), neoplasm, or arteriovenous malformation, or
 - e. Clinical history, past imaging, or clinical judgement suggests that the intracranial occlusion is chronic.

Etiology of Neurologic Symptoms

2. Non-ischemic or non-thrombotic etiology of current stroke symptoms such as suspected cerebral vasospasm, infectious source (e.g., bacterial endocarditis, septic shock), complications from acute drug abuse (e.g., cocaine intoxication).

Timing of Endovascular Thrombectomy (EVT)

3. Availability of endovascular thrombectomy (EVT) for current AIS (in whom EVT is indicated) where planned EVT would be feasible (e.g., time of femoral artery puncture), within 1 hour following the start of study drug infusion.

Medical History and Current Health Status***Bleeding Risks***

4. Medical history or active clinically significant bleeding, lesions, or conditions (at the investigator's judgement) considered to be of significant risk for major bleeding; (this may include but not limited to any history of intracranial hemorrhage or vascular aneurysm of the large arteries of major intraspinal or intracerebral abnormalities).
5. Cerebral infarction within 90 days of screening.
6. Arterial dissection involving any intracranial artery or the aortic arch.
7. Confirmed acute coronary syndrome within 90 days of screening.
8. Severe hepatic impairment as defined by decompensated liver disease (e.g., Child-Pugh Class C) or clinically significant hepatic condition associated with coagulopathy or bleeding risk.
9. Medical history of chronic renal failure, any condition requiring dialysis or renal replacement therapy or evidence of acute kidney injury at the time of screening, an estimated glomerular filtration rate $< 60 \text{ mL/min/1.73m}^2$.
10. Severe, uncontrolled hypertension (systolic blood pressure $\geq 185 \text{ mmHg}$ or diastolic blood pressure $\geq 110 \text{ mmHg}$) that cannot be controlled with antihypertensive therapy.
11. Known bleeding diathesis (hereditary or acquired) or any significant coagulopathy. Specifically, platelet count $< 100,000/\mu\text{L}$, international normalized ratio > 1.7 , aPTT > 40 seconds, or prothrombin time > 15 seconds.
12. Major trauma, surgery, or invasive procedures:
 - a. Severe head trauma within 3 months of screening or acute head trauma.
 - b. Major trauma not involving the head within 14 days of screening.
 - c. Major surgery with 14 days of screening.
 - d. Intracranial or intraspinal surgery within 90 days of screening.

- e. Dural puncture (e.g., lumbar puncture) or arterial puncture of a noncompressible blood vessel within 7 days of screening.

General

- 13. Pre-existing medical, neurological, or psychiatric disease that would confound the neurological or functional evaluations of this study.
- 14. Pre-treatment blood glucose > 400 mg/dL (22.20 mmol/L) or Pre-treatment blood glucose < 50 mg/dL (2.78 mmol/L) unless it is corrected prior to study treatment administration. Participants with subsequently normalized blood glucose levels may be considered for inclusion, per Investigator judgement.
- 15. Known hereditary or acquired abnormality of UGT1A1 metabolism or deficiency such as Gilbert's syndrome (hereditary liver condition with elevated bilirubin), Crigler-Najjar syndrome (UGT1A1 gene defect associated with congenital non-hemolytic jaundice).
- 16. Prolonged QT with QTc of > 450 ms for males and > 460 ms for females.
- 17. Life expectancy less than 6 months due to comorbid condition.

Medications

- 18. Prior thrombolytic administration within 90 days of screening or planned thrombolytic administration for the AIS. (Co-administration with any other thrombolytic agent(s) within 48 hours is prohibited).
- 19. Known or suspected use of any oral anticoagulant therapy, including but not limited to vitamin K antagonists, thrombin inhibitors, or factor Xa inhibitors, within 48 hours of screening unless blood testing confirms absence of drug substance in the system.
- 20. Use of dual anti-platelet therapy, IV aspirin, or glycoprotein IIb/IIIa inhibitors within 24 hours of screening. Pre-screening use of oral antiplatelet monotherapy with aspirin at doses less than or equal to 325 mg daily OR clopidogrel 75 mg daily is permitted.
- 21. Use of heparin or low molecular weight heparin at a therapeutic dose, excluding prescreening prophylactic dose low molecular weight heparin with a normal aPTT.
- 22. Use of nephrotoxic medications or agents within 7 days of screening that would (in the investigator's opinion) pose significant risk of acute kidney injury (e.g., aminoglycosides, cyclosporins, lactams, amphotericin B, cisplatin, indomethacin, etc.).

Other Exclusion Criteria

23. Prior treatment with JX10, any known history of systemic hypersensitivity reaction or anaphylaxis to JX10, the excipients contained in the formulation, and if applicable, any diagnostic agents anticipated to be administered during the study.
24. Participants who are pregnant or currently breastfeeding, and those intending to become pregnant during the study.
25. Current enrollment or a plan to enroll in another investigational device study, or interventional clinical study in which an investigational treatment or approved therapy for investigational use is administered within 90 days prior to the screening.
26. Substance abuse history including heavy alcohol use over the past year, per Investigator judgement, or acute intoxication with alcohol or other substance that may impair participant's ability to comply with treatment and assessment.
27. Inability to comply with study requirements or other unspecified reasons that, in the opinion of the Investigator or Sponsor, make the participant unsuitable for enrollment.

5.3. Screening, Retesting, and Screen Failures**5.3.1. Screening**

Once informed consent is obtained, screening assessments can occur. At this time, a unique identification number is assigned that will be used on study-related documents pertaining to the participant. Any identification numbers that are assigned will not be reused even if the participant does not receive treatment or continue in the study. Study sites are required to document all screened participants initially considered for inclusion in the study.

Participants, or their legally authorized representative (LAR), or physician per local requirements must provide informed consent before any screening tests are performed (see Section 10.4). Determination of whether consent by a LAR is required (and if so, determination of the LAR) as well as specific details of the consenting process will be determined by country law, state law, and local ethics committee requirements. The study team is encouraged to use fax, telephone, and/or telehealth consent if allowed by country law, state law, and local ethics committee rules. Participating study sites are required to document all screened candidates initially considered for inclusion in the study.

Because pertinent participant information will already be collected as part of standard of care, and to reduce additional study-specific procedures, baseline information may be taken from the participant's medical records prior to obtaining informed consent. However, informed consent will be obtained prior to performing any study-specific procedures.

Baseline NIHSS, pre-stroke mRS, vital signs, imaging, 12-lead ECG, and other assessments collected as standard of care for the presenting AIS event may be utilized for eligibility.

5.3.2. Retesting

Participants who fail screening criteria may be eligible for inclusion if repeat CT/CTA or CTP imaging is performed by the Investigator per standard of care, with all other inclusion criteria being met. No extension to the 24-hour screening period may be granted. Participants who have an out-of-range laboratory result may be retested at the discretion of the Investigator. Participants may also be rescreened if NIHSS is the original reason for the screen failure (i.e., presenting NIHSS is < 5 , but neurologic deterioration occurs within the screening period and all other entry criteria are met). Blood pressure may be repeated as needed per Investigator judgement.

5.3.3. Screen Failures

Screen failures are defined as participants who sign the ICF but are not subsequently randomized. Participating study sites are required to document all screened candidates initially considered for inclusion in this study. If a participant is considered a screen failure, the reasons for exclusion must be documented in the participant's source documents. A minimal set of screen failure information is required to ensure transparent reporting of screen failure participants to meet the Consolidated Standards of Reporting Trials publishing requirements and to respond to queries from regulatory authorities. Minimal information includes demography, screen failure details, eligibility criteria, and any SAE. Rescreening of participants who are screen failures is at the discretion of the Sponsor; see the operations manual for additional information.

5.4. Randomization

Randomization will be performed using interactive response technology (IRT). Participants will be randomized when all screening assessments have been completed and after the Investigator has verified that the participants are eligible per criteria in Section 5.1 and Section 5.2. Participants will be assigned a unique identification number that will be used on study-related documents pertaining to the participant. Any participant identification numbers that are assigned will not be reused even if the participant does not receive treatment or continue in the study.

In Part 1, all participants will be assigned randomly in a 1:1:1 ratio to receive JX10 at dosing levels of 1 mg/kg, 3 mg/kg, or placebo. In Part 2, all participants will be assigned randomly in a 1:1 ratio to receive either JX10 (optimal dose selected after the end of Part 1 analysis) or placebo.

As part of randomization, stratification factors will be used in Part 1 and Part 2 to minimize treatment group imbalance over the following baseline covariates: stroke type (M1 or proximal M2 or ICA vs. others), NIHSS assessed at the time of screening (> 10 vs. ≤ 10), and stroke onset time (4.5 to 6 hours vs. > 6 to 12 hours vs. > 12 hours). A central randomization list will be

created using stratified block randomization. The detailed algorithm will be documented in the randomization specification.

6. STUDY TREATMENT

6.1. Regimen

The study drug (either JX-10 or placebo) will be administered over 31 minutes (an IV bolus of 10% of the total dose over 1 minute, followed by a continuous IV infusion of the remaining 90% of the total dose over 30 minutes). Study treatment should be initiated and completed as soon as possible after randomization. During Part 1, participants will receive a single dose of either JX10 1 mg/kg, 3 mg/kg (up to a maximum dose of 100 and 300 mg, respectively), or placebo IV. During Part 2, participants will receive either JX10 or placebo.

The infusion of study treatment through a dedicated peripheral IV line is preferred. A calibrated infusion pump with a dedicated infusion line will be used to ensure infusion at the specified rates. The study treatment may not be delivered through a central line or peripherally inserted central catheter line. No other medication may be administered in the same line as the study treatment, nor should the line be used for blood withdrawal.

The date and time of the start and end of the infusion will be recorded in the eCRF. The start and stop times for interruption and the reason for interruption should be recorded.

6.1.1. Modification of Dose and/or Treatment Schedule

The dosage cannot be modified.

6.2. Study Treatment Management

Study treatment will be manufactured, handled, and stored in accordance with applicable Good Manufacturing Practices.

Site staff should follow the pharmacy manual, for specific instructions on the handling, preparation, administration, and disposal of the study treatment. The pharmacy manual aligns with all other references, including the protocol.

Study treatment must be dispensed only by a pharmacist or appropriately qualified staff. Study treatment is to be dispensed only to participants enrolled in this study. Once study treatment is prepared for a participant, it can be administered only to that participant. Study treatment is for one-time use only; do not use any study treatment remaining in the vial for another participant.

6.2.1. Clinical Formulation

JX10 is manufactured in accordance with GMPs. JX10 API is supplied as a sterile lyophilized drug product in a single-use clear glass vial shrink-wrapped in a yellow transparent film. Each vial contains JX10 equivalent to 100 mg of JX10 free base, HCO-60, meglumine, sodium hydroxide, and hydrochloric acid.

The placebo is also supplied as a sterile lyophilized product in a single-use clear glass vial shrink-wrapped in a yellow transparent film containing HCO-60, meglumine, sodium hydroxide, and hydrochloric acid.

6.2.2. Stability, Storage, and Handling Instructions

To ensure that the most up-to-date procedures are followed when storing or preparing JX10 for clinical use, please follow the instructions provided in the JX10 pharmacy manual.

JX10 is to be protected from light and stored at 2°C to 8°C (36°F to 46°F), in a monitored, locked refrigerator, with limited access.

The individual preparing JX10 should carefully review the instructions provided in the pharmacy manual. JX10 is to be administered via an IV bolus followed by an IV infusion.

6.3. Blinding Procedures

The Investigator, study staff, and participants will remain blinded to the treatment assignments. To maintain the study blind, it is imperative that treatment assignments are not shared with the participants, their families, or any member of the blinded study team, both at the site and at the Sponsor and CRO, wherever applicable. To ensure that efficacy and outcome assessors are blinded to study treatment, evaluators performing all efficacy and outcome evaluations will be blinded to knowledge of AEs and SAEs during the acute hospitalization phase of the study (should not have participated in the treatment during the acute hospitalization phase of the study nor review the participant's hospital records from the acute hospitalization study period).

Participants randomized to the placebo arm will be allocated by the IRT system to match the number of vials and reconstitution volume to mirror the active treatment arms.

Upon study completion, the Sponsor will provide the randomization codes to Investigators, who then can inform their participants about the treatment received.

In the event that an emergency unblinding of a participant's treatment assignment is required, refer to Section [8.3.4.4.1](#).

6.4. Guidance on Care of Participants: Precautions, Monitoring, and Treatment

6.4.1. General Considerations

All participants receiving study treatment should, in general, at minimum, receive similar management as other post-thrombolysis AIS patients per local standard of care, local guidance, and Investigator judgement, including admission to an inpatient unit with expertise in managing and monitoring AIS patients [Powers 2019]. The guidance here serves as minimum recommendations and standards for all participants.

All participants should be monitored closely for signs and symptoms of hemorrhage. The placement of nasogastric tubes, indwelling catheters, or intra-arterial pressure catheters should be delayed for the first 4 hours after study treatment if the participant can be safely managed without them, per Investigator judgement.

Participants should be monitored for neurological deterioration, especially sudden or abrupt changes (See Sections 6.4.3 and 6.4.4). Neurological deterioration is defined as neurological changes that are 4 points or more on the NIHSS from baseline, or from the lowest NIHSS between baseline and 24 hours (See also Section 8.3.3). For frequency of assessments, refer to Section 6.4.3.

For information on concomitant therapies, see Section 6.6.

6.4.2. Blood Pressure and Neurological Monitoring

Blood pressure should be maintained at < 180/105 mm Hg for at least the first 24 hours after administration of study treatment.

Measure BP and perform neurological assessments every 15 minutes ($\pm$ 5 minutes) during and after the JX10 administration for 2 hours, then every 30 minutes ($\pm$ 10 minutes) for 6 hours, then hourly ($\pm$ 15 minutes) until 24 hours after study treatment.

Increase the frequency of BP measurements, per Investigator judgement, if SBP is > 180 mmHg or if DBP is > 105 mmHg. Administer antihypertensive medication to maintain BP at or below these levels, per Investigator judgement.

Monitor for signs or symptoms of suspected ICH (See 6.4.3 for details). Discontinue infusion and obtain an emergency CT scan if the patient develops severe headache, acute hypertension, nausea, or vomiting, or has a worsening neurologic examination.

Confirm neurological deterioration by repeating full NIHSS (deterioration is $\geq$ 4 points). Arrange a CT immediately if there is neurological deterioration.

In neurologic deterioration unrelated to ICH, institute continued monitoring and supportive. In cases of uncontrolled cerebral edema, emergent neurosurgical support may be necessary.

6.4.3. Intracranial Hemorrhage or Suspected Intracranial Hemorrhage

If a participant develops signs or symptoms suggestive of intracranial hemorrhage (e.g., severe headache, acute hypertension, nausea, vomiting, or worsening neurological examination), then immediately discontinue the infusion of study treatment and obtain an emergency head CT. Management of any intracranial hemorrhage should be initiated, as appropriate per Investigator clinical judgement or institutional guidelines. In addition to emergency imaging, hematology labs, such as CBC, fibrinogen, and clotting testing and fluid and blood product support should be instituted. Hematology and Neurosurgery consultation may be required. In serious cases, fibrinogen concentrate, or cryoprecipitate and platelets may be administered, if platelet count $< 100 \times 10^9/L$.

If no intracranial hemorrhage is detected and resumption of study drug administration is deemed safe per the Investigator's judgement, then the Investigator may consider restarting the suspended infusion of study treatment, as long as the interruption is not greater than 15 minutes (as the half-life of the study treatment is approximately 15 minutes). In most cases, resumption of the infusion would likely not be feasible. See Section 6.5 for more details.

6.4.4. Clinically Significant or Major Bleeding

If the participant develops a clinically significant hemorrhage (per Investigator judgement) or major bleeding (as defined in Section 8.3.3.3), then immediately discontinue study treatment infusion. Management of hemorrhage should be initiated, as appropriate including by not limited to treating the underlying lesion, fluid resuscitation, and possible blood transfusion.

6.4.5. Anaphylaxis and Hypersensitivity Reactions

Participants should be closely monitored for anaphylaxis and other hypersensitivity reactions. Medications and resuscitation equipment for the emergency management of anaphylactic reactions and hypersensitivity reactions must be available in the room where the infusion of study treatment is being performed.

6.5. Compliance

Compliance with treatment dosing is to be monitored and recorded by the designated site staff.

Study treatment should not be suspended at any time during the infusion unless necessary for management of an AE or suspected AE. Any interruption of study treatment should be recorded on the eCRF along with a reason. If dose interruption (single or multiple) is greater than 15 minutes in total, then study treatment administration should be discontinued.

Study treatment will be administered by the site staff according to pharmacy manual.

6.6. Concomitant Therapy and Procedures

6.6.1. Concomitant Therapy

A concomitant therapy is any drug or substance administered between the participant's consent and last study visit.

Participants should be instructed to contact their Investigators before taking any new medications, including non-prescription drugs and herbal preparations.

6.6.2. Prohibited Concomitant Therapy

JX10 is not intended for use in combination with any drug with fibrinolytic action. As such, use of any other thrombolytic is prohibited for a minimum of 48 hours following administration of study treatment. Patients with prior exposure of a thrombolytic within 90 days of screening are not eligible for the study.

In general, antithrombotic medications including antiplatelet drugs and anticoagulants (including at prophylactic doses) should be avoided for the first 24 hours following administration of study treatment. The 24-hour NCCT head or MRI brain should be obtained prior to initiation of any antithrombotic medication after administration of study treatment to ensure that there is no intracranial hemorrhage.

If cases where the antithrombotic treatment such as that used (e.g., heparin) during endovascular thrombectomy is critical, diligent care and dosing must be exercised to minimize risk of hemorrhage. For low molecular weight heparin, the administered dose should not exceed the maximum recommended dose for prophylactic use in the specific country. Pre-screening use of oral antiplatelet monotherapy with aspirin at doses less than or equal to 325 mg daily OR clopidogrel 75 mg daily is permitted.

6.6.3. Concomitant Procedures

A concomitant procedure is any therapeutic intervention (e.g., surgery/biopsy, physical therapy) or diagnostic assessment (e.g., blood gas measurement, bacterial cultures) performed between the time the participant is enrolled in the study (consent) and the last study visit.

6.7. Continuation of Treatment

There is no provision to provide study treatment after the study, as study treatment is intended for single use only.

7. DISCONTINUATION AND WITHDRAWAL OF PARTICIPANTS FROM THE STUDY

7.1. Discontinuation of Study Treatment

A participant must permanently discontinue study treatment for any of the following reasons:

- The participant experiences an intracranial hemorrhage or other clinically significant hemorrhage (per Investigator judgement).
- Anaphylaxis or other hypersensitivity reaction related to JX10 or its excipients (per Investigator judgement).
- The participant experiences an AE that necessitates permanent discontinuation of the study treatment.
- The participant experiences a medical emergency that necessitates unblinding of the participant's treatment assignment.

The primary reason for discontinuation of study treatment must be recorded in the participant's eCRF.

In the event that an AE or medical emergency temporarily interrupts the study treatment, study treatment may resume at the discretion of the study investigator.

Participants who discontinue study treatment should remain in the study and continue protocol-required tests and assessments.

7.2. Lost to Follow-Up

Participants will be considered as potential lost to follow-up if they repeatedly fail to return for scheduled visits and are unable to be contacted by the study site. Every attempt should be sought to reach the participant even if successful contact is not established until after 90 days; at the minimum, vital status should be ascertained. Every attempt to contact the participant should be documented. Participants are not considered permanently lost until the end of the study.

Given the life-altering event of AIS and need of care and rehabilitation, the participant's living arrangements may change which include temporary or permanent placement at residential care or nursing homes. Alternate contact information including that of LAR should be obtained prior to hospital discharge and confirmed via phone or other means of communication post discharge. Permission to contact other parties such as primary care providers, authorized family members or friends, social worker, etc., should be obtained, if applicable.

The following actions must be taken if a participant fails to return to the study site for a required study visit:

- The site must attempt to contact the participant and reschedule the missed visit as soon as possible and counsel the participant on the importance of maintaining the assigned visit schedule and ascertain whether the participant wishes to and/or should continue in the study.
- Contact designated alternate contact such as LAR.
- Certified mail to participant and/or authorized designated alternate contact such as LAR.
- Contact the site monitor for further instructions and assistance.

In cases in which the participant is deemed lost to follow-up, the investigator or designee must make every effort to regain contact with the participant. These contact attempts should be documented in the participant's medical record. Work with your site monitor for further instructions and assistance. If all attempts fail (depending on local regulation), publicly available information such as death registries or other registries may be accessed or private investigation to locate the participant may be initiated (where permitted).

7.3. Withdrawal of Participants from the Study

Participants must be withdrawn from the study for any one of the following reasons:

- The participant enrolls into another interventional clinical study in which an investigational treatment or approved therapy for investigational use is administered.
- The participant is unwilling or unable to comply with the protocol.
- The participant is identified as being pregnant after study enrollment but did not receive study treatment. If a participant receives study treatment and is later identified as being pregnant, then the study treatment should be immediately discontinued but the participant should remain in the study for scheduled assessments and safety follow up per protocol.
- The participant/LAR withdraws consent to continue study treatment. Participant withdrawal may occur any time the participant or LAR wishes to no longer continue with the study. Every attempt must be made to obtain information about the reason(s) for discontinuation, and any possible AEs.
- Other reason at the discretion of the Investigator or at the discretion of the Sponsor.

The primary reason for the participant's withdrawal from the study must be recorded in the participant's eCRF. Samples and data collected from subject may be reviewed in support of study analyses and to comply with study requirements and laws.

Participants should undergo an early termination visit unless withdrawal is due to death or withdrawal of consent.

Participants who withdraw from the study may not be replaced; however, additional recruitment may be performed (at the discretion of Sponsor) to supplement these participants. Details about missing data handling is provided in Section 9.6.

Early termination is defined as withdrawal from the study. Participants who discontinue study treatment for any reason or who move to comfort care/palliative care should not be withdrawn from the study. Rather, they should continue protocol-required tests and assessments.

In this population, participants are neurologically compromised, often requiring transitional care, such as skill nursing facilities or other arrangement. Other complications precipitated by the AIS such as falls, respiratory illnesses, mobility issues, etc. may all complicate follow-up and affect the participant's ability to attend visit or complete all the assessments. Before withdrawal, in addition to discussion with the patient and LAR, options for an abridgement assessment, telehealth evaluation, visiting nurse, etc. should be considered. Discuss with your site monitor for instructions and assistance. Once last contact at Day 90 visit with minimum assessment of mRS should be encouraged or at minimum once last contact to assess vital status. Options of withdrawal and the process of informed withdrawal will be provided in the operations manual.

8. TRIAL ASSESSMENTS AND PROCEDURES

8.1. Efficacy, Pharmacokinetic, and Pharmacodynamic Assessments

See [Table 1](#) for the timing of all assessments.

Tests and evaluations affecting primary endpoints and/or analyses may need to be repeated if the original results are lost or damaged. In these cases, participants will be asked to return to the study site to have the evaluations repeated. As best practice, the clinical assessments should be performed by the same qualified individual. Investigational sites should ensure qualified personnel perform screening scales such as mRS and NIHSS and maintain documentation of proficiency. Further training and instructions can be found in the operations manual. The recommended sequence of assessments as well as additional processes and procedures are detailed in the operations manual.

8.1.1. Clinical Efficacy Assessments

The following clinical assessments will be performed to evaluate the efficacy of JX10:

- mRS Score
- Overall mortality rate
- Revascularization/reperfusion on follow-up CTA/MRA

- Penumbral salvage (reversal of $T_{\max} > 6$ seconds lesion) using baseline CTP/MR DWI, PWI, and 24-hour NCCT or MRI
- Final infarct volume assessment on MRI DWI or on NCCT
- NIHSS Score
- Barthel Index Score (BIS)
- Stroke Impact Scale – 16 (SIS-16)
- EuroQol 5-level assessment of health outcomes (EQ-5D-5L)
- mTICI Score on angiography (if performed)

8.1.2. Laboratory Target Engagement and Pharmacodynamic Assessments

The following laboratory assessments will be performed to evaluate the target engagement and pharmacodynamics of JX10:

- Blood samples will be collected for biomarker assessments (protein) and coagulation assessments.

8.1.3. Pharmacokinetic Assessments

The following parameters will be calculated to assess the pharmacokinetics of JX10:

- The concentrations of JX10 in plasma.

PK assessments require diligent preparation as well as sample collection and handling. Multiple samplings can be invaluable in spite of the logistical challenges. Investigational sites and patients will be given the option to be included in the more frequent PK sampling. The goal is to obtain an adequate number of samples to help inform dose-finding and further future development of the compound. Further instructions will be provided in the operations manual.

In general, for participants who have provided consent to PK sampling, PK samples will be collected at the timepoints specified in [Table 1](#) (Schedule of Activities for Part 1 and Part 2), i.e., at:

1. pre-dose,
2. 0 hour post-dose/end of infusion + 5 minutes,
3. 1 hour post-dose/end of infusion $\pm$ 5 minutes, and
4. 24 hours post-dose-/end of infusion $\pm$ 2 hours.

PK assessments should be done as close to these timepoints as possible, and the windows are only provided for operational reference.

To collect adequate data from participants who are administered with JX10 for the potential population PK analysis, PK samples will be collected from approximately 90 participants in Part 1, and from approximately 120 participants in Part 2.

8.1.4. Pharmacodynamic Assessments

The pharmacodynamic properties of JX10 will be assessed through biomarker assessments that will include, but may not be limited to, plasma or serum levels of the following:

- DHET
- EET
- PAP
- alpha-2 antiplasmin

8.1.5. Pharmacogenetic and Genetic Assessments

Not applicable.

8.2. Safety Assessments

See [Table 1](#) for the timing of all safety assessments.

8.2.1. Clinical Safety Assessments

The following clinical assessments will be performed to evaluate the overall safety profile of JX10, but not limited to:

- AE and SAE recording.
- Medical history.
- Physical examination: focused system assessment with heart and lung examination and any additional assessment related to complaints or AE.
- Neurologic examination: in addition to general assessment, there should be a focus assessment related to change in neurological status related to stroke signs and symptoms.

- Vital sign assessments: temperature, pulse rate, systolic and diastolic blood pressure, oxygen saturation, and respiratory rate. Vital signs may be taken with the participant in the sitting, recumbent/semi-recumbent, or supine position per Investigator discretion.
- 12-lead electrocardiograms (ECGs): ECGs should be taken while the participant is recumbent/semi-recumbent or supine (as per Investigator's discretion).
- Continuous Holter/ECG monitoring with ECG extraction for centralized extraction and analysis will be performed where feasible in a select number of participants to enable Concentration-QTc (C-QTc) analysis. ECGs will be extracted from a Holter monitor for C-QTc analysis. Refer to Section 9.5.5 for the definition of C-QTc analysis set. Further instructions will be provided in a separate operations manual.
- Continuous cardiac telemetry for at least 24 hours following administration of study treatment.
- Neuroimaging.
- Laboratory safety assessments.
- Brief suicidal ideation and behavior assessment.
- Concomitant therapy and procedure recording.

8.2.2. Laboratory Safety Assessments

The following laboratory assessments will be performed locally to evaluate the safety profile of JX10:

- Hematology: Complete blood count with differential, absolute neutrophil count, and platelet count.
- Blood chemistry: total protein, albumin, creatinine, eGFR, blood urea nitrogen (where possible), uric acid (where possible), bilirubin (total and direct), alkaline phosphatase, glucose, calcium, phosphorus, bicarbonate (where possible) or CO₂ (where possible), chloride, sodium, and potassium.
- Liver function: alanine aminotransferase, aspartate aminotransferase, gamma glutamyl transferase, total bilirubin, indirect bilirubin.
- Coagulation: INR, PT, aPTT, fibrinogen.
- Pregnancy testing (urine/serum).

- Urinalysis: specific gravity, pH, macroscopic evaluation for blood, protein, and glucose. Perform microscopic examination for applicable abnormalities. Urinalysis is applicable to participants who can provide a urine sample spontaneously or, if planned, via catheter placement. *Test for myoglobin in the event that the urinalysis is heme positive.*
- Urine albumin-to-creatinine ratio (UACR), urine protein-to-creatinine ratio (UPCR), creatinine kinase, LDH, and haptoglobin.

8.2.3. Product-Specific Safety Assessments

Not applicable.

8.3. Safety Definitions, Recording, Reporting, And Responsibilities

Throughout the course of the study, every effort must be made to remain alert to possible AEs. If an AE occurs, the first concern should be for the safety of the participant. If necessary, appropriate medical intervention should be provided.

At the signing of the ICF, each participant or his/her LAR and/or main caregiver must be given the names and telephone numbers of site staff for safety event reporting, pregnancies, and medical emergencies, etc. Throughout the protocol, the Sponsor is named, but reporting may be done through a CRO.

8.3.1. Definitions

8.3.1.1. Adverse Event

An AE is any untoward medical occurrence in a participant or clinical investigation subject (participant) administered a pharmaceutical product and that does not necessarily have a causal relationship with this treatment. An AE can therefore be any unfavorable and unintended sign (including an abnormal assessment such as an abnormal laboratory value), symptom, or disease temporally associated with the use of a medicinal (investigational) product, whether or not related to the medicinal (investigational) product.

AEs include:

- Exacerbation of a chronic or intermittent pre-existing condition including either an increase in frequency and/or intensity of the condition.
- New conditions detected or diagnosed after IP administration even though it may have been present before the start of the trial.
- Abnormal diagnostic assessments, (e.g., change on physical examination, ECG findings), if they represent a clinically significant finding that was not present at trial start or worsened during the trial.

- Laboratory test abnormalities if they represent a clinically significant finding, symptomatic or not, which was not present at trial start or worsened during the trial, require treatment or led to dose reduction, interruption or permanent discontinuation of IP. In general, abnormal laboratory findings without clinical significance (based on the investigator's judgment) are not to be recorded as AEs. Where applicable, clinical sequelae (not the laboratory abnormality) are to be recorded as the AE. Abnormal laboratory test results or other abnormal findings that were present or found at baseline but have not worsened than expected should also not be recorded as AEs or SAEs.
- Signs, symptoms, or the clinical sequelae of a suspected drug-drug interaction.
- Signs, symptoms, or the clinical sequelae of a suspected overdose of either IP or a concomitant medication. Overdose per se will not be reported as an AE/SAE unless it is an intentional overdose taken with possible suicidal/self-harming intent. Such overdoses should be reported regardless of sequelae.

Generally, determination of whether an abnormal assessment (e.g., laboratory value, vital sign, and electrocardiogram) result meets the definition of an AE will be made by the Investigator. Abnormal results are not considered AEs unless one or more of the following criteria are met:

- The result meets the criteria for an SAE.
- The result requires the participant to receive specific corrective therapy or modification of treatment.
- The result is considered by the Investigator to be clinically significant.

The same clinically significant laboratory abnormality or abnormal vital sign found in multiple follow-ups should not be repeatedly recorded as an AE or SAE unless there are changes in the severity or etiology.

8.3.1.2. Serious Adverse Event

An SAE is any untoward medical occurrence that at any dose:

- Results in death.
- Is life-threatening (i.e., the adverse event, in the view of the investigator, places the participant at immediate risk of death). It does not refer to an include an event that, had it occurred in a more severe form, might have caused death.
- Requires inpatient hospitalization or prolongation of existing hospitalization.
- Results in persistent or significant disability/incapacity.

- Results in a congenital anomaly/birth defect.
- Is a medically important event.

A medically important event is an AE that, in the opinion of the Investigator, may jeopardize the participant or may require intervention to prevent one of the other outcomes listed in the definition above. (Examples of such medical events include allergic bronchospasm requiring intensive treatment in an emergency room or convulsions occurring at home that do not require an inpatient hospitalization.)

8.3.1.3. Prescheduled or Elective Procedures or Routinely Scheduled Treatments

A prescheduled or elective procedure or a routinely scheduled treatment will not be considered an SAE, even if the participant is hospitalized. The study site must document all the following:

- The prescheduled or elective procedure or routinely scheduled treatment was scheduled (or was on a waiting list to be scheduled) prior to obtaining the participant's consent to be in the study.
- The condition requiring the prescheduled or elective procedure or routinely scheduled treatment was present before and did not worsen or progress in the opinion of the Investigator between the participant's consent to be in the study and the time of the procedure or treatment.
- The prescheduled or elective procedure or routinely scheduled treatment is the sole reason for the intervention or hospital admission.
- If a participant is hospitalized due to local requirements for administration of the study treatment, the hospitalization should not be considered an SAE unless one of the requirements in Section 8.3.1.2 is met.

8.3.2. Safety Classifications

8.3.2.1. Investigator Assessment of Events

All events must be assessed to determine the following:

- If the event meets the criteria for an SAE as defined in Section 8.3.1.2.
- The relationship of the event to study treatment as defined in Section 8.3.2.2.
- The severity of the event as defined in Section 8.3.2.3.

The terms "severe" and "serious" are not synonymous. Severity refers to the intensity of an adverse event (e.g., rated as mild, moderate, or severe, see Section 8.3.2.3); the event itself may

be of relatively minor medical significance (such as severe headache without any further findings). Severity and seriousness need to be independently assessed for each adverse event recorded on the eCRF.

8.3.2.2. Relationship of Events to Study Treatment

Each AE must be assessed by the investigator, based on clinical judgment, as to whether or not there is a reasonable possibility of causal relationship to the IP and reported as either related or unrelated. A “reasonable possibility” of a relationship conveys that there are facts, evidence, and/or arguments to suggest a causal relationship, rather than a relationship cannot be ruled out. Alternative causes, such as underlying disease(s), concomitant therapy, and other risk factors, as well as the temporal relationship of the event to IP administration will be considered and investigated.

Relationship of Event to Study Treatment	
<i>The following definitions should be considered when evaluating the relationship of AEs and SAEs to the study treatment.</i>	
Not related	An AE will be considered “not related” to the use of the investigational product if there is not a reasonable possibility that the event has been caused by the product under investigation. Factors pointing toward this assessment include but are not limited to the lack of reasonable temporal relationship between administration of the investigational product and the AE, the presence of a biologically implausible relationship between the product and the AE, or the presence of a more likely alternative explanation for the AE.
Related	An AE will be considered “related” to the use of the investigational product if there is a reasonable possibility that the event may have been caused by the product under investigation. Factors that point toward this assessment include but are not limited to a positive rechallenge, a reasonable temporal sequence between administration of the investigational product and the AE, a known response pattern of the suspected product, improvement following discontinuation or dose reduction, a biologically plausible relationship between the product and the AE, or a lack of an alternative explanation for the AE.

For each AE/SAE, the investigator must document in the medical notes that he/she has reviewed the AE/SAE and has provided an assessment of causality. There may be situations in which an SAE has occurred, and the investigator has minimal information to include in the initial report to sponsor. However, it is very important that the investigator always assesses causality for every event before the initial transmission of the SAE data to sponsor.

The investigator may change his/her opinion of causality considering follow-up information and send an SAE follow-up report with the updated causality assessment.

8.3.2.3. Severity of Events

The following definitions should be considered when evaluating the severity of AEs:

Severity of Event	
Mild	Symptom(s) barely noticeable to subject or does not make subject uncomfortable; does not influence performance or functioning; prescription drug not ordinarily needed for relief of symptom(s) but may be given because of personality of subject.
Moderate	Symptom(s) of a sufficient severity to make subject uncomfortable; performance of daily activity is influenced; subject is able to continue in study; treatment for symptom(s) may be needed.
Severe	Symptom(s) causes severe discomfort; symptoms cause incapacitation or significant impact on subject's daily life; severity may cause cessation of treatment with study treatment; treatment for symptom(s) may be given and/or subject hospitalized.

A mild, moderate, or severe AE may or may not be serious. Medical judgment should be used on a case-by-case basis. Seriousness, rather than severity assessment, determines the regulatory reporting obligations.

8.3.2.4. Expectedness of Events

Expectedness of all SAEs will be determined by the Sponsor and the JX10 Investigator's Brochure will serve as the reference safety document.

8.3.3. Monitoring and Recording Events

8.3.3.1. Adverse Events

Any AE experienced by the participant between the time of the signing of the ICF and the last Follow-up visit is to be recorded on the eCRF, regardless of the severity of the event or its relationship to study treatment. At each study visit, the Investigator will assess the participant for AEs and will record any new AEs or updates to previously reported AEs on the eCRF.

AEs that are ongoing when the participant completes or discontinues the study will be followed by the Investigator until the event has resolved, stabilized, or returned to baseline status. AE outcome will not be recorded on the eCRF, as applicable.

A treatment-emergent adverse event is defined as any new AE reported or any worsening of an existing condition on or after initiation of study treatment.

8.3.3.2. Adverse Events of Special Interest

An AE of special interest is an AE of scientific and medical concern specific to this study, for which ongoing monitoring and reporting are required.

The following adverse events are considered adverse events of special interest and will be entered on the AE eCRF and reported to the Sponsor on the Adverse Events of Special Interest (AESI) Form within 24 hours of the site staff becoming aware of the event:

- Major Hemorrhage, consisting of intracranial hemorrhage and major bleeding
- Hemolysis
- Rhabdomyolysis
- Acute Kidney Injury

Serious and non-serious AESIs will be recorded on the AE eCRF and reported on the Clinical Trial AESI Form per the guidelines in Section 8.3.3.5.

8.3.3.3. Definition of Specific Hemorrhage and Cerebral Edema Events

Adverse events of intracranial hemorrhages will be classified as symptomatic or asymptomatic. Symptomatic intracranial hemorrhage is defined as local or remote parenchymal hemorrhage type 2, subarachnoid hemorrhage, and/or intraventricular hemorrhage within 36 hours post-randomization, combined with a neurological deterioration of 4 points or more on the NIHSS from baseline, or from the lowest NIHSS value between baseline and 24 hours, or leading to death [Rao 2014].

Adverse events of hemorrhage will be classified as major bleeding or non-major bleeding, according to the International Society on Thrombosis and Haemostasis definition.

Major bleeding is defined as

1. fatal bleeding, and/or
2. symptomatic bleeding in a critical area or organ, and/or
3. a reduction in the hemoglobin level of at least 2 g/dL (1.24 mmol/L) or more, or leading to transfusion of 2 or more units of whole blood or red cells [Rodeghiero 2010].

Adverse events of cerebral edema will be classified as symptomatic or asymptomatic. Symptomatic cerebral edema is defined as brain edema with mass effect as the predominant cause of clinical deterioration [Hacke 2008].

8.3.3.4. Serious Adverse Events

Any SAE experienced by the participant between the time of the signing of the ICF and the last Follow-up visit is to be recorded on an SAE form, regardless of the severity of the event or its relationship to study treatment. Thereafter, the event should be reported to the Sponsor only if the Investigator considers the SAE to be related to study treatment.

SAEs must be reported to the Sponsor within 24 hours (as described in Section 8.3.6.1) or according to national law. Follow-up information regarding an SAE also must be reported within 24 hours or according to the national law.

Any SAE that is ongoing when the participant completes or discontinues the study will be followed by the Investigator until the event has resolved, stabilized, or returned to baseline status. If the Investigator becomes aware of any SAE with a suspected causal relationship to the investigational medical product that occurs after the end of the clinical trial in a subject treated by him or her, the Investigator will, without undue delay, report the SAE to the sponsor.

All events of symptomatic intracranial hemorrhage will be designated as serious adverse events.

8.3.3.5. Immediate Reporting of Serious Adverse Events

In order to adhere to all applicable laws and regulations for reporting an SAE, the study site must formally notify the Sponsor or designee within 24 hours of the site staff becoming aware of the SAE or according to national law. It is the Investigator's responsibility to ensure that the SAE reporting information and procedures are used and followed appropriately.

Reporting Information for SAEs

A report **must be submitted** to the Sponsor regardless of the following:

- Whether or not the participant has undergone study-related procedures
- Whether or not the participant has received study treatment
- The severity of the event
- The causal relationship of the event to study treatment

To report initial or follow-up information on an SAE, fax or email a completed SAE form to the Sponsor or its designee; refer to the operations manual for complete contact information.

8.3.3.5.1. Deaths

Death is an outcome of an event. The event that resulted in death should be recorded on the appropriate eCRF. All causes of death must be reported as SAEs within 24 hours of the site becoming aware of the event or according to national law. The Investigator should make every

effort to obtain and send death certificates and autopsy reports to the Sponsor. The term death should be reported as an SAE only if the cause of death is not known and cannot be determined.

8.3.3.5.2. Suspected Unexpected Serious Adverse Reactions

Suspected unexpected serious adverse reactions (SUSARs) are SAEs that are unexpected and judged by the Investigator or the Sponsor to be related to the study treatment administered.

Appropriate personnel at the Sponsor will unblind SUSARs for the purpose of regulatory reporting. The Sponsor will submit SUSARs (in blinded or unblinded fashion) to regulatory agencies according to local law. The Sponsor will submit SUSARs to Investigators in a blinded fashion.

8.3.4. Procedures for Handling Special Situations

8.3.4.1. Pregnancy

Female participants should not become pregnant during the study and for at least 30 days after their dose of study treatment. Male participants should not impregnate their partners during the study and for at least 90 days after their dose of study treatment.

The Investigator must report a pregnancy occurring in a female participant and/or female partner of a male participant from first dose of study treatment to 90 days by faxing or emailing the appropriate form to the Sponsor within 24 hours of the site staff becoming aware of the pregnancy. Refer to the operations manual's Official Study Contact List for complete contact information. The Investigator or site staff must report the outcome of the pregnancy to the Sponsor. A pregnancy is not considered an AE and should not be recorded on the AE eCRF.

Congenital abnormalities and birth defects in the offspring of male or female participants should be reported as an SAE if conception occurred during the study treatment period or within 90 days from their last dose of study treatment.

8.3.4.2. Overdose

An overdose is any dose of study treatment administered to a participant or taken by a participant that exceeds the dose assigned to the participant according to the protocol. Overdoses are not considered AEs and should not be recorded as an AE on the eCRF; however, all overdoses must be recorded on an Overdose form and faxed or emailed to the Sponsor within 24 hours of the site becoming aware of the overdose. An overdose must be reported to the Sponsor even if the overdose does not result in an AE. If an overdose results in an AE, the AE must be recorded. If an overdose results in an SAE, both the SAE and Overdose forms must be completed and faxed or emailed to the Sponsor. All study treatment-related dosing information must be recorded on the dosing eCRF.

8.3.4.3. Abuse, Misuse, Medication Error, Accidental or Occupational Exposure

Abuse, misuse, medication error, and accidental or occupational exposure events are not considered AEs and should not be recorded as an AE on the eCRF; however, all occurrences of abuse, misuse, medication error, and accidental or occupational exposure with the study treatment administered must be recorded on a Clinical Trial Special Situation(s) Report Form and faxed or emailed to the Sponsor within 24 hours of the site becoming aware of the event. If an abuse, misuse, medication error, or accidental or occupational exposure event results in an AE, the AE must be recorded on the eCRF. If an abuse, misuse, medication error, or accidental or occupational exposure event results in an SAE, both the SAE and the Clinical Trial Special Situation(s) Report Forms must be completed and faxed or emailed to the Sponsor within 24 hours of the site becoming aware of the events. All study treatment-related dosing information must be recorded on the dosing eCRF.

8.3.4.3.1. Abuse

Abuse is when the persistent or sporadic, intentional excessive use of the study treatment which is accompanied by harmful physical or psychological effects.

8.3.4.3.2. Misuse

Misuse is when the medicinal product is intentionally and inappropriately used not in accordance with the authorized/approved product information.

8.3.4.3.3. Medication Error

Medication error is any preventable incident that may cause or lead to inappropriate study treatment use or participant harm while the study treatment is in the control of the health care professionals or participant. Such an incident may be due to health care professional practice, product labeling, packaging and preparation, procedures for administration, and systems, including the following: prescribing, order communication, nomenclature, compounding, dispensing, distribution, administration, education, monitoring, and use.

8.3.4.3.4. Accidental or Occupational Exposure

Accidental or occupational exposure is the unintentional exposure to a study treatment as a result of one's professional or non-professional occupation, or accidental exposure to a non-professional to whom exposure was not intended (i.e., study product given to wrong participant).

8.3.4.4. Medical Emergency

In a medical emergency requiring immediate attention, site staff will apply appropriate medical intervention, according to current standards of care. The Investigator (or designee) should

contact the study's Medical Director. Refer to the operations manual's Official Study Contact List for complete contact information.

8.3.4.4.1. Unblinding for Medical Emergency

In a medical emergency when knowledge of the participant's treatment assignment may influence the participant's clinical care, the Investigator may access the participant's treatment assignment by IRT. The Investigator must document the reasons for unblinding in the participant's source documents. The Investigator is strongly advised not to divulge the participant's treatment assignment to any individual not directly involved in managing the medical emergency, or to personnel involved with the analysis and conduct of the study. The Investigator can contact the Sponsor to discuss such situations.

8.3.5. Contraception Requirements

All women of childbearing potential must have a negative pregnancy test prior to randomization and agree to avoid becoming pregnant during the study and ensure that effective contraception is used during the study and for at least 30 days after their last dose of study treatment. Fertile male participants whose partners are females of childbearing potential must agree to use 2 effective contraceptive methods, both of which may be barrier methods to prevent pregnancy with their partner(s), during the study and up to 90 days after their last dose of study treatment. In addition, participants should not donate sperm or eggs for the duration of the study and for at least 90 days after their last dose of study treatment.

Female participants and female partners of fertile male participants must be:

- willing to remain abstinent (not engage in sexual intercourse with an opposite sex partner for the duration of the screening period and throughout the study, and for 30 days after the last dose), or
- willing to use 2 effective methods of birth control throughout the study and for 30 days after the last dose.

For the purposes of this study, women of childbearing potential are defined as all women physiologically capable of becoming pregnant, UNLESS they meet one of the following conditions:

- Postmenopausal
 - 52 continuous weeks of natural (spontaneous) amenorrhea without an alternative medical cause and a serum follicle-stimulating hormone (FSH) level > 40 mIU/mL
 - 6 weeks after surgical bilateral oophorectomy with or without hysterectomy
- Post hysterectomy

- Female surgical sterilization (e.g., bilateral tubal ligation) where applicable according to local guidelines

For the purposes of the study, effective contraception is defined as use of one of the following:

For females:

- Established, defined as at least 3 months, use of oral, intravaginal, or transdermal combined (estrogen and progestogen containing) hormonal methods of contraception associated with the inhibition of ovulation (e.g., combined oral contraceptives, patch, vaginal ring, injectables, and implants). If a participant discontinues use of this method during the study, then the participant must elect an alternative method of effective contraception and inform the Investigator.
- Established, defined as at least 3 months, use of oral, injected, or implanted progestogen-only hormonal methods of contraception associated with the inhibition of ovulation. If a participant discontinues use of this method during the study, then the participant must elect an alternative method of effective contraception and inform the Investigator.
- Placement of an intrauterine device or intrauterine hormone-releasing system.
- Barrier methods of contraception where applicable according to local guidelines (e.g., female condom, cervical cap, diaphragm).
- Bilateral tubal occlusion.
- Sex with a male who has undergone surgical sterilization (with the appropriate post vasectomy documentation of the absence of sperm in the ejaculate).

For males:

- Vasectomy with negative semen analysis at follow-up. If documentation is not available, the participant must use contraception.
- Condoms with spermicide where applicable according to local guidelines.
- Sex with a woman who uses the methods described for females if she is of childbearing potential.

True abstinence, when this is consistent with the preferred and usual lifestyle of the participant, can be considered an acceptable method of contraception based on the evaluation by the Investigator who should also take into consideration the duration of the clinical trial. Periodic abstinence (e.g., calendar, ovulation, symptothermal, post-ovulation methods) and withdrawal are not considered acceptable methods of contraception.

Pregnancy reporting is described in Section 8.3.4.1.

8.3.6. Safety Responsibilities

8.3.6.1. The Investigator

The Investigator's responsibilities include the following:

- Monitor and record all AEs, including SAEs and AESIs, on the eCRF regardless of the severity or relationship to study treatment.
- Determine the seriousness, relationship, and severity of each event.
- Determine the onset and resolution dates of each event.
- Monitor and record all pregnancies in female participants and/or female partners of a male participants and follow up on the outcome of all pregnancies.
- Complete an SAE form for each SAE and fax or email it to the Sponsor within 24 hours of the site staff becoming aware of the event or according to national law.
- Complete an AESI form for each AESI and fax or email it to the Sponsor within 24 hours of the site staff becoming aware of the event or according to national law.
- Pursue SAE and AESI follow-up information actively and persistently. Follow-up information must be reported to the Sponsor within 24 hours of the site staff becoming aware of new information or according to national law.
- Complete an Overdose Form and/or Special Situations Form for each event as defined in Sections 8.3.4.2 and 8.3.4.3 and fax or email to the Sponsor within 24 hours of the site staff becoming aware of the event.
- Ensure all AE and SAE reports are supported by documentation in the participants' medical records.
- Pursue AE follow-up information, if possible, until the event has resolved or become stable. Record AE follow-up information, including resolution, on the eCRF, as applicable.
- Report SAEs to local ethics committees, as required by local law.
- The investigator is obligated to perform or arrange for the conduct of supplemental measurements and/or evaluations as medically indicated or as requested by sponsor to elucidate the nature and/or causality of the AE or SAE as fully as possible. This may

include additional laboratory tests or investigations, histopathological examinations, or consultation with other health care professionals.

- If a participant dies during participation in the trial or during a protocol-defined follow-up period, the investigator will provide sponsor with a copy of any post-mortem findings including histopathology if it has been performed.

8.3.6.2. The Sponsor

The Sponsor's responsibilities include the following:

- Before a site can enroll any participants, the Medical Monitor is responsible for reviewing with site staff the definitions of AE, SAEs, and special situations, as well as the instructions for monitoring, recording, and reporting AEs, SAEs, and special situations.
- The Sponsor is to notify all appropriate regulatory authorities, ethics committees/IRBs, and Investigators of SAEs including SUSARs, as required by local law, regulatory authorities, investigators/institutions, and ethics committees/IRBs, within required time frames (e.g., U.S. Food and Drug Administration, EudraVigilance).

9. STATISTICAL METHODS AND DETERMINATION OF SAMPLE SIZE

The objectives of the study and the endpoints to be analyzed are listed in Section 3.

9.1. General Considerations

A summary of analysis methods for primary and secondary efficacy endpoints, pharmacokinetic endpoints, pharmacodynamic endpoint, and safety endpoints for Part 1 and Part 2 is provided in this Section. More details about analysis methods of tertiary endpoints, missing data handling, and other supportive analyses will be provided in the SAP, which will contain the final details on the statistical methods used in this study.

Part 1 data will be locked after the last participant enrolled in Part 1 completes the study or terminate study early. When Part 1 is completed, radiographic, clinical, safety, and PK data will be analyzed to support the dose selection decision. Part 1 participants and data will not be included in the Part 2 analysis.

If the statistical model fails to converge, the corresponding endpoint will be re-analyzed with further adjustment in the statistical analysis model. Alternative approaches may include removing the covariates which cause the non-convergence issue from the statistical analysis model. More details will be provided in the SAP.

9.2. Statistical Hypotheses

No formal statistical hypothesis testing will be performed in part 1.

In part 2, the analyses evaluating treatment effect on the primary and secondary efficacy endpoints will test the null hypothesis that there is no treatment difference between patients receiving placebo and those receiving JX10 in the Full Analysis Set (FAS). Adjustments for multiplicity will be specified in Section 9.4.

9.3. Sample Size Considerations

In Part 1, each JX10 dose group (1 mg/kg, 3 mg/kg) will be compared to the placebo group for proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days. Odds ratio of improvement on proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days comparing each of the JX10 dose groups to the placebo group will be estimated. An odds ratio greater than 1 means JX10 is favored over placebo for this outcome. At a 1-sided significance level of 0.05, 80 participants per arm will provide more than 90% power when the true odds ratio is 3.0 for a JX10 dose group compared to the placebo group and when the proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days is assumed to be 18.4% for placebo (based on observations from the Japanese Phase 2a clinical experience). No multiplicity adjustment will be made.

In Part 2, the odds ratio of improvement on proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days comparing JX10 to the placebo will be estimated. An odds ratio greater than 1 means JX10 is favored over placebo for this outcome. Assuming the odds ratio of JX10 to placebo is 2.0, the proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days in placebo group is assumed to be 18.4%, the 2-sided type I error rate as 0.05, a sample size of 500 (approximately 250 randomized to JX10 and 250 randomized to placebo) provides 91% power to detect the difference in proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days between JX10 and placebo. The sample size of Part 2 might be modified based on the interim analysis result of Part 1 if there is significant deviation between the actual interim analysis result from the estimated parameter values to maintain the intended power.

9.4. Multiplicity Adjustment

No multiplicity adjustment will be made for Part 1 since Part 1 data will not be included in the Part 2 primary endpoint analysis.

In Part 2, the null hypothesis for the primary and secondary efficacy variables in the FAS will be tested in the pre-specified order using a closed testing procedure. For the preservation of the overall type I error rate at two-sided 0.05, the primary and secondary endpoints will be tested in the following specified order using a closed testing procedure. If the primary endpoint achieves statistical significance at two-sided 0.05, then secondary endpoints will be tested with two-sided

0.05, with their testing in a pre-specified sequential order as listed in Section 3. The detailed description will be provided in the Statistical Analysis Plan (SAP).

9.5. Analysis Sets

The following definitions describe the populations used for specific study analysis:

9.5.1. All Randomized Set

All Randomized Set includes all participants who are randomized.

9.5.2. Full Analysis Set

In Part 1 and Part 2, Full Analysis Set (FAS) is defined as those participants randomized and received at least one dose of the study treatment (JX10 or placebo). Baseline characteristics and efficacy analysis will be conducted based on the FAS. The efficacy analyses will be conducted based on the planned treatment randomized.

9.5.3. Safety Analysis Set

In Part 1 and Part 2, safety analysis set is defined as all participants who are enrolled in Part 1 and Part 2 of the study and have received at least one dose of the study treatment (JX10 or placebo). The safety analyses will be conducted based on the actual treatment received. Safety endpoints will be summarized as descriptive statistics. No statistical hypothesis testing will be performed on safety endpoints.

The safety analysis will be conducted based on the safety analysis set.

9.5.4. Pharmacokinetic/Pharmacodynamic Analysis Set

In Part 1 and Part 2, Pharmacokinetic/Pharmacodynamic analysis set (PKPDS) is defined as all participants who receive JX10 and have at least one post-baseline plasma PK and PD measurement.

The PK and PD analysis will be conducted based on the PKPD analysis set.

9.5.5. Concentration-QTc Analysis Set

Concentration-QTc Analysis Set (C-QTcS) is defined as the participants who have, received JX10, have at least one postbaseline plasma PK measurement and the corresponding QTc data, and the participants who receive placebo and have at least one QTc data.

9.6. Statistical Analysis

9.6.1. Analysis for Efficacy Endpoints

The efficacy analysis will be conducted based on the FAS. Participants will be analyzed based on the intention-to-treat principle, according to their randomized treatment assignment regardless of treatment received.

In Part 1, pairwise comparison of each JX10 dose group to the placebo group will be used to evaluate the treatment effect of different JX10 dose levels. No multiplicity adjustment is planned in Part 1 of the study.

In Part 2, all tests will be conducted at 1-sided significant level of 0.025 (unless otherwise specified). To control type 1 error rate, if the primary endpoint is significant at the 1-sided significant level of 0.025, the secondary endpoints will be tested at the same significance level in a sequential order to claim the treatment effect of JX10 on the endpoint is statistically significant compared to placebo. See Section 9.4 for the details of the multiplicity adjustment.

Additionally, a 1-sided P-value less than 0.05 will be considered as providing evidence of benefit of JX10 compared to placebo warranting further study of JX10.

9.6.1.1. Analysis of the Primary Endpoint

Analysis of primary endpoint will be performed on the FAS.

The primary endpoint of the study is proportion of participants with no or minimal symptoms (mRS score 0-1) at 90 days. Logistic regression will be used to estimate an odds ratio of improvement on the Day 90 mRS dichotomized at 0-1 vs. 2-6. The treatment effect estimations will be derived using logistic regression models that include baseline covariates, namely, stroke type (M1 or proximal M2 or ICA vs. others), NIHSS assessed at the time of screening (> 10 vs. ≤ 10), and stroke onset time (4.5 to 6 hours vs. > 6 to 12 hours vs. > 12 hours). The Odds ratio and 95% CI will be calculated to evaluate the treatment effect and will be estimated for JX10 compared to the placebo group.

The primary analysis estimand for the primary endpoint with intercurrent events (ICEs) and strategies for addressing ICEs are listed in Table 3. The number of participants with missing values due to reasons other than death is expected to be small. Death is not considered as an ICE if the endpoint is mRS. Any cause of death will be recorded as 6 for mRS outcomes for all the subsequent scheduled study visits. Sensitivity analyses will be performed to evaluate the impact of missing data due to reasons other than death, and the missing data handling will be specified in the SAP. Other sensitivity analyses for the primary endpoint and the details will be included in the SAP.

9.6.1.2. Analysis of the Secondary Endpoint

Analysis of secondary endpoints will be performed on the FAS.

The secondary endpoints are listed below:

- Day 90 ordinal mRS score, based on a 6-point ordinal scale (0, 1, 2, 3, 4, 5/6).

Ordinal logistic regression will be used and it will be analyzed using the proportional odds model to estimate an odds ratio of improvement on the Day 90 mRS as a 6-point ordinal scale (0, 1, 2, 3, 4, 5/6), adjusting for the baseline stratification factors defined in Section 5.4. The proportional odds model will be used in the analysis.

- Proportion of participants with functional independence (mRS score 0-2) at 90 days.

Logistic regression will be used to estimate an odds ratio of improvement on the Day 90 mRS dichotomized at 0-2 vs. 3-6, adjusting for the baseline stratification factors defined in Section 5.4.

- Proportion of participants with BIS > 90 at 90 days.

Logistic regression will be used to estimate an odds ratio of improvement on the Day 90 BIS dichotomized at > 90 vs. ≤ 90, adjusting for the baseline stratification factors defined in Section 5.4.

- Proportion of participants with favorable outcome (mRS 0-1 or 0-2), taking into account baseline NIHSS score (mRS sliding dichotomy).

Logistic regression will be used to estimate an odds ratio of improvement on the Day 90 favorable outcome (mRS 0-1 or 0-2) taking into account baseline NIHSS score (mRS sliding dichotomy) Yes vs No, adjusting for the baseline stratification factors defined in Section 5.4. The sliding dichotomy approach is adopted by customizing the definition of a favorable outcome (mRS 0-1 or 0-2) to each participant's baseline NIHSS score. For the baseline NIHSS score ≤ 10, mRS 0-1 will be defined as the favorable outcome, while for baseline NIHSS score > 10, mRS 0-2 will be defined as the favorable outcome.

- Proportion of participants with reperfusion at 24 hours post-treatment, defined as > 90% reduction in $T_{\max}$ > 6 seconds lesion volume or complete recanalization on CT angiography (CTA)/magnetic resonance angiography (MRA)*.
- Proportion of participants with recanalization at 24 hours post-treatment, defined as complete or partial recanalization on CT angiography (CTA)/magnetic resonance angiography (MRA)*.

- Proportion of participants with angiographic reperfusion (TICI 2b/3 or TICI 3) at completion of angiographic procedure (endovascular thrombectomy participants only)*.
- Proportion of participants where planned thrombectomy was not performed (planned endovascular thrombectomy participants only)*.

*Logistic regression model after adjustment using baseline covariates and stratification factors will be used to estimate the odds ratios of these respective endpoints.

- Proportion of participants with improvement on the NIHSS by > 7 points or score 0 or 1 at 7 days or hospital discharge.

Logistic regression will be used to estimate an odds ratio of improvement at 7 days or hospital discharge NIHSS compared to the baseline measurement (the closest collection before administration of the study treatment), dichotomized at > 7 points or having a score of 0 or 1 at 7 days vs. ≤ 7 , adjusting for the baseline stratification factors defined in Section 5.4.

Table 3: Estimand of Efficacy Endpoint Analysis

Estimand	Analysis Population	Variable	Intercurrent Events (ICEs)	Strategy for Addressing ICEs	Measurement of interest
Primary Estimand	FAS population	The Day 90 mRS results as binary endpoint: 0-1 vs. 2-6	EVT	Treatment Policy	All FAS participants will be included in the analysis regardless of whether they underwent EVT or not. The analysis is to use logistic regression model.
		The Day 90 mRS results as binary endpoint: 0-2 vs. 3-6	Carotid endarterectomy and carotid stenting procedure	Treatment Policy	All FAS participants will be included in the analysis regardless of whether they received the procedure or not. The analysis is to use the logistic regression model.
		Proportion of participants with favorable outcome (mRS 0-1 or 0-2), taking into account baseline NIHSS score (mRS sliding dichotomy)	Initiation of other treatment including rehabilitation or physiotherapy	Treatment Policy	All FAS participants will be included in the analysis regardless of whether they received other treatment or not. The analysis is to use the logistic regression model.
			Discontinuation of study treatment	Treatment Policy	All FAS participants will be included in the analysis regardless of the volume of study treatment they received. The analysis is to use the logistic regression model.
			Lost to follow-up other than death before Day 90	Hypothetical	Missing data will be considered as missing at random (MAR) and will be imputed using multiple imputation under the assumption of MAR.
Primary Estimand	FAS population	The 7 days or hospital discharge NIHSS	EVT	Treatment Policy	All FAS participants will be included in the analysis regardless of whether they underwent EVT or not. The analysis is to use logistic regression model and

Estimand	Analysis Population	Variable	Intercurrent Events (ICEs)	Strategy for Addressing ICEs	Measurement of interest
		improvement as binary endpoint: > 7 or score 0 or 1 vs. ≤ 7			analysis of variance for continuous endpoints.
			Discontinuation of study treatment	Treatment Policy	All FAS participants will be included in the analysis regardless of the volume of study treatment they received. The analysis is to use the logistic regression model for binary endpoints and analysis of variance for continuous endpoints.
			Lost to follow-up other than death before 24 hours	Hypothetical	Missing data will be considered as missing at random (MAR) and will be imputed using multiple imputation under the assumption of MAR.
			Death before 24 hours	Composite Variable	Death will be considered as non-responder.
Primary Estimand	FAS population	The Day 90 mRS results in 6-category	EVT	Treatment Policy	All FAS participants will be included in the analysis regardless of whether they underwent EVT or not using proportional odds model.
			Carotid endarterectomy and carotid stenting procedure	Treatment Policy	All FAS participants will be included in the analysis regardless of whether they received the procedure or not. The analysis is to use the proportional odds model.
			Initiation of other treatment including rehabilitation or physiotherapy	Treatment Policy	All FAS participants will be included in the analysis regardless of whether they received other treatment or not. The analysis is to use the proportional odds model.
			Discontinuation of study treatment	Treatment Policy	All FAS participants will be included in the analysis regardless of the volume of study treatment they received. The analysis is to use the proportional odds model.
			Lost to follow-up other than death before Day 90	Hypothetical	Missing data will be considered as missing at random (MAR) and will be imputed using multiple imputation under the assumption of MAR.
Primary Estimand	FAS population	The Day 90 BIS as binary endpoint: > 90 vs. ≤ 90	EVT	Treatment Policy	All FAS participants will be included in the analysis regardless of whether they underwent EVT or not. The analysis is to use logistic regression model.
			Carotid endarterectomy and carotid stenting procedure	Treatment Policy	All FAS participants will be included in the analysis regardless of whether they received the procedure or not. The analysis is to use the logistic regression model.

Estimand	Analysis Population	Variable	Intercurrent Events (ICEs)	Strategy for Addressing ICEs	Measurement of interest
			Initiation of other treatment including rehabilitation or physiotherapy	Treatment Policy	All FAS participants will be included in the analysis regardless of whether they received other treatment or not. The analysis is to use the logistic regression model.
			Discontinuation of study treatment	Treatment Policy	All FAS participants will be included in the analysis regardless of the volume of study treatment they received. The analysis is to use the logistic regression model.
			Lost to follow-up other than death before Day 90	Hypothetical	Missing data will be considered as missing at random (MAR) and will be imputed using multiple imputation under the assumption of MAR.
			Death before Day 90	Composite Variable	Death will be considered as non-responder.
Primary Estimand	FAS population	Proportion of participants with reperfusion at 24 hours post-treatment	Discontinuation of study treatment	Treatment Policy	All FAS participants will be included in the analysis regardless of the volume of study treatment they received. The odds ratios for each JX10 dose compared to placebo will be calculated by using logistic regression model.
		Proportion of participants with recanalization at 24 hours post-treatment	Death before angiographic evaluation	Composite Variable	Death will be considered as non-responder.
Primary Estimand	FAS population, subset the population as appropriate	Proportion of participants with angiographic reperfusion (TICI 2b/3 or TICI 3) at completion of angiographic procedure	Discontinuation of study treatment	Treatment Policy	All FAS participants will be included in the analysis regardless of the volume of study treatment they received. The odds ratios for each JX10 dose compared to placebo will be calculated by using logistic regression model.
		Proportion of participants where planned thrombectomy was not performed	Death before planned thrombectomy	Composite Variable	Death will be considered as non-responder.

9.6.1.3. Analysis of the Tertiary Endpoints

The tertiary endpoints will be summarized based on the type of endpoints. Continuous endpoints will be summarized using descriptive statistics (number of participants [n], mean, SD, median, Q1, Q3, minimum, and maximum). Categorical endpoints will be summarized using frequency

and percentage of participants. No multiplicity adjustment is planned. Further details will be specified in the SAP.

9.6.2. Analysis for Pharmacokinetic Endpoints

Both Part 1 and Part 2 of the study will collect only sparse PK samples; thus, JX10 concentrations in plasma will be summarized by timepoint and dose group using descriptive statistics.

Population PK analysis may be conducted to estimate PK parameters and to identify the impact of potential covariates on the variability of JX10 PK.

Exposure-response analysis may also be conducted to help select the dose for Part 2.

The results from population PK analysis and exposure-response analysis may be included in a separate report.

9.6.3. Analysis for Pharmacodynamic Endpoints

Plasma biomarkers are tertiary endpoints. Summary statistics for actual values and change from baseline will be presented by treatment group.

9.6.4. Analysis for Safety Endpoints

Safety endpoints will be summarized based on the safety analysis set of each part of the study as defined in Section 9.5.3. Participants will be analyzed based on the actual treatment received. No statistical hypothesis testing will be performed on safety endpoints.

A central facility will be selected by the Sponsor to read and interpret all imaging data for safety evaluation.

9.6.4.1. Adverse Events

AEs will be coded using MedDRA.

A treatment-emergent adverse event is defined as follows:

- Onset occurred any time after the start of study treatment, and/or
- Event worsened since it was previously reported (including worsening of signs, symptoms, laboratory values, or diagnoses that were present prior to the first dose of study treatment but then worsened any time after the start of study treatment)

The incidence of AEs will be summarized as follows:

- By MedDRA preferred term

- By primary SOC
- By primary SOC and MedDRA preferred term
- By severity
- By relationship to study treatment

The incidence of SAEs will also be summarized by primary SOC and MedDRA preferred term.

For the summary of AEs by severity, if a participant experiences an event more than once at different severities during the study, then that participant will be counted only once under the maximum severity within each SOC/preferred term.

For incidence of relationship to study treatment, a participant will be counted only once and in the category of the strongest relationship to study treatment within each SOC/preferred term. The relationship to study treatment will be classified as related or not related.

Any deaths that occurred during the study will be listed. Listings will be presented by participant for all AEs as well as for SAEs, AEs associated with death, and AEs leading to discontinuation of study treatment or study withdrawal.

9.6.4.2. Clinical Laboratory Results

Clinical laboratory evaluations include hematology, blood chemistry, liver function tests, coagulation, pregnancy testing, and urinalysis. Laboratory data will be summarized using shift tables. The number and percentage of participants with shifts from baseline to high or low status for hematology and blood chemistry and shifts from baseline to high or positive status for urinalysis will be presented by treatment group. Descriptive statistics for actual values and changes from baseline will also be summarized by treatment group.

9.6.4.3. Vital Signs

Vital sign measures include temperature, heart rate, systolic and diastolic blood pressure, oxygen saturation, and respiratory rate. The analysis of vital signs will focus on clinically relevant abnormalities. Descriptive statistics for actual values and changes from baseline for vital signs will be presented by treatment group. A separate summary will be presented for incidence of clinically relevant postbaseline abnormalities. The definitions of abnormalities for temperature, heart rate, systolic and diastolic blood pressure, oxygen saturation, and respiratory rate values will be provided in the SAP.

9.6.4.4. 12-Lead ECGs

Actual values and changes from baseline for the 12-lead ECGs will be presented using descriptive statistics by treatment group. A separate summary will be presented for incidence of clinically relevant abnormalities at baseline and postbaseline.

9.6.4.5. Physical and Neurologic Examinations

Exams will be designated by the Investigator as normal, abnormal not an AE, or abnormal and an AE.

9.6.4.6. Tertiary Pharmacodynamic Biomarkers

Descriptive statistics will be used to summarize the tertiary PD biomarker levels and change from baseline by visit.

9.6.5. Patient Disposition

The number of patients who are randomized, who complete the planned treatment, and who prematurely discontinue from the study will be presented by treatment group and overall. Reasons for premature discontinuation will also be summarized.

9.6.6. Demographics and Other Baseline Characteristics

Patient demographics and other baseline characteristics will be summarized descriptively by treatment group.

9.6.7. Concomitant Medications

Concomitant medications will be summarized by drug class and preferred term using the World Health Organization (WHO) Drug Dictionary.

9.6.8. Interim Analysis

At minimum, there will be one planned interim analysis in Part 1 and 2, respectively. The Part 1 interim analysis will be performed when the last participant in Part 1 has been enrolled. The primary goal of this analysis is to determine the dose to advance into Part 2. The overall benefit-risk will be assessed based on the results of efficacy, safety, and tertiary endpoints to select an optimal JX10 dose for Part 2. No multiplicity adjustment is needed in the dose selection analysis since there is no formal hypothesis testing.

The dose selection analysis will be conducted by the unblinded biostatistician or independent statistical team, working with the IDMC. Sites and participants will remain blinded to the Part 1 data until Part 2 is completed. Details of the management of unblinding will be prespecified and documented in the Unblinding Plan. The interim analysis plan and timing in Part 2 will be defined upon completion of Part 1 and will be pre-specified in the Part 2 SAP.

9.7. Subgroup Analysis

To determine whether the treatment effect is consistent across various subgroups, the estimate of the treatment effect (with a nominal 95% CI) for the primary endpoint will be estimated within each category of the following classification variables:

- Age (> 65 , ≤ 65 years)
- Sex (Female, Male)
- Race
- Geographic region
- Stroke type (M1 or proximal M2 or ICA vs. others)
- NIHSS assessed at the time of screening (> 10 vs. ≤ 10)
- Stroke onset time (4.5 to 6 hours vs. > 6 to 12 hours vs. > 12 hours)

The consistency of the treatment effect will be assessed descriptively via a summary statistics by category for the classification variables listed above.

10. ETHICAL AND REGULATORY REQUIREMENTS

The Sponsor, any contracted third party, and the Investigator must comply with all instructions, regulations, and agreements in this protocol and applicable International Council for Harmonisation (ICH) and Good Clinical Practice (GCP) guidelines and conduct the study according to local regulations. The clinical trial is to be conducted in compliance with the Regulation (EU) No 536/2014. Data will be collected and processed in accordance with the General Data Protection Regulation (GDPR) EU 2016/679.

The Investigators are responsible for demonstrating timely oversight of all clinical trial data from their site, including data external to the electronic data capture system, such as laboratory, imaging, and electronic clinical outcomes assessment data. Investigators must approve all their data on completed CRFs by signing electronically, at the subject, visit, or casebook level, at any time prior to an interim lock or database lock, as well as before any subsequent re-lock. The electronic data capture system does not prohibit Investigator approval or signing in any way.

The Investigator may delegate responsibilities for study-related tasks where appropriate to individuals sufficiently qualified by education, training, and experience, in accordance with applicable ICH and GCP guidelines. The Investigator should maintain a list of the appropriately qualified persons to whom significant study-related duties have been delegated. The Investigator

is responsible for supervising those individuals and for implementing procedures to ensure the integrity of the tasks performed and any data generated.

The Investigator shall report any data breaches that might occur to the Sponsor or its designee, without undue delay. The Sponsor has implemented a business practice to address data breaches that complies with the requirements of applicable laws and regulations including the General Data Protection Regulation (GDPR). The data breach procedures in the business practice provide specific responses to actual or potential threats and involve investigation, containment and mitigation. If applicable, the authorities and the data subjects shall be notified of data breach, within the required timeframes of the applicable laws and regulations, including those of the GDPR.

10.1. Declaration of Helsinki

This study will be performed in alignment with the ethical principles outlined in the Declaration of Helsinki.

10.2. Ethics Committee

The Investigator must obtain ethics committee approval of the protocol, ICF, and other required study documents prior to starting the study. The Sponsor or designee may submit documents on behalf of the study sites in countries other than the United States.

If the Investigator makes any changes to the ICF, the Sponsor must approve the changes before the ICF is submitted to the ethics committee. A copy of the approved ICF must be provided to the Sponsor. After approval, the ICF must not be altered without the agreement of the relevant ethics committee and the Sponsor.

It is the responsibility of the Investigators to ensure that all aspects of institutional review are conducted in accordance with current applicable regulations.

The Sponsor must receive a letter documenting ethics committee approval, which specifically identifies the protocol, protocol number, and ICF, prior to the initiation of the study. Protocol amendments will be subject to the same requirements as the original protocol.

A progress report must be submitted to the ethics committee at required intervals and not less than annually.

At the completion or termination of the study, where required, the study site must submit a close-out letter to the ethics committee and the Sponsor.

10.3. Changes to Final Protocol

All protocol amendments must be submitted to the ethics committee and regulatory authorities if required by local law. Protocol modifications that affect participant safety, the scope of the

investigation, or the scientific quality of the study must be approved by the ethics committee before implementation of such modifications to the conduct of the study. If required by local law, such modifications must also be approved by the appropriate regulatory agency prior to implementation.

However, the Sponsor may, at any time, amend this protocol to eliminate an apparent immediate hazard to a participant. In this case, the appropriate regulatory authorities will be notified subsequent to the modification.

In the event of a protocol modification, the ICF may require similar modifications (see Section 10.4).

10.4. Informed Consent

Prior to performing any study-related activities under this protocol, including screening tests and assessments, informed consent with the approved ICF must be obtained, for each part of this study per applicable country regulations. In the event a participant is unable to consent for him/herself, the method of consent with an alternative person allowed to consent (e.g., relative, caregiver, the participant's legally authorized representative) either in person or remotely, including the use of eConsent, should adhere to local regulations. Likewise, the use of emergency consent signed by an independent physician, study investigator, or any other relevant party may be implemented if permitted and in compliance with local regulations.

The background of the proposed study, the procedures, the benefits and risks of the study, and that study participation is voluntary for the participant must be explained to the participant or the participant's legally authorized representative. The participant or the participant's legally authorized representative must be given sufficient time to consider whether to participate in the study.

If the original consenting process involved anyone other than the participant, and if required by local standards and regulations, consent will be sought for the remaining procedures from the participant once they are deemed to have regained capacity.

Participants will be informed that their race and ethnicity will be collected during the study (unless the collection is not permitted by applicable law or not approved by the governing ethics committee) and the data will be used during analysis of study results.

A copy of the signed and dated ICF must be given to the participant or the participant's legally authorized representative. The original signed and dated ICF will be retained with the study records. Local regulations must be complied with in respect to the final disposition of the original and copies of the signed and dated ICFs.

Confirmation of informed consent must also be documented in the participant's medical record.

When additional information that may affect participants' willingness to continue in the study becomes available, the Investigators will be notified in a timely manner, according to all local and applicable law. An updated ICF may be required.

10.5. Participant Data Protection

Prior to any testing under this protocol, including screening tests and assessments, candidates must also provide all authorizations required by applicable national and local privacy regulations (e.g., Protected Health Information authorization in North America).

During the study, participants' race and ethnicity will be collected (unless the collection is not permitted by applicable law or not approved by the governing ethics committee). These data will be used in the analysis of the safety and/or pharmacokinetic profile of the study treatment. Collection of these data will further the understanding of underlying disease mechanisms and the response to treatment.

Study reports will be used for research purposes only. The participant will not be identified by name in CRFs, study-related forms, study reports, or any related publications. The Sponsor, its partners and designees, ethics committees, and various government health agencies may inspect the records of this study. Every effort will be made to keep the participant's personal medical data confidential.

10.6. Compensation for Injury

The Sponsor maintains appropriate insurance coverage for clinical studies and will follow applicable local compensation laws.

10.7. Conflict of Interest

The Investigators should address any potential conflicts of interest (e.g., financial interest in the Sponsor) with the participant before the participant makes a decision to participate in the study.

10.8. Study Report Signatory

The Sponsor will follow all applicable local regulations pertaining to study report signatories.

10.9. Registration of Study and Disclosure of Study Results

The Sponsor will register the study and post study results regardless of outcome on a publicly accessible website in accordance with the applicable laws and regulations.

The Sponsor also will notify, when required, the regulatory authorities and ethics committees about the completion or termination of this study and send a copy of the study synopsis in accordance with necessary timelines.

10.10. Retention of Study Data

The study samples may be stored up to 25 years after the end of the study or per applicable regulations. All the collected samples, study images, and ECG recordings will be de-identified, coded, and handled per ICH guidelines. Future research will only be limited to the IMP.

The minimum retention time for study records will meet the strictest standard applicable to that site, as dictated by any institutional requirements or local, national, or regional laws or regulations. Prior to proceeding with destruction of records, the Investigator must notify the Sponsor in writing and receive written authorization from the Sponsor to destroy study records. In addition, the Investigator must notify the Sponsor of any changes in the archival arrangements including but not limited to archival at an offsite facility or transfer of ownership if the Investigator leaves the site.

11. KEY ROLES AND STUDY COMMITTEES

11.1. Site Staff

The Investigator of the site will designate the appropriate site personnel and their roles in the study. See the Study Reference Guide for details.

The primary treating neurologist will be responsible for the following:

- Management of the routine neurological care of the participant.
- Assessment (including assignment of causality) and treatment of AEs.
- Review of hematology and all blood chemistry results to assess whether the participant's study treatment should be temporarily withheld or permanently discontinued.
- Providing education to participants on signs and symptoms of potential delayed reactions after treatment and providing information to the participant on how to contact the site to report any AE.

The treating neurologist may designate other medical personnel (i.e., the backup treating neurologist or the treating nurse) at the investigational site to perform some of the tests and evaluations listed under "treating neurologist." If there is more than 1 treating neurologist available at a given site such that each one is assigned to participants, then these treating neurologists may act as backup for each other. The same holds true for the treating nurses and examining technicians.

For a given participant, all sites should attempt to maintain the same mRS rater throughout the study.

11.2. Vendors

The Sponsor will ensure oversight of any study-related duties and functions carried out on its behalf and will specify in writing all duties and functions that are transferred.

11.2.1. Contract Research Organization

A CRO will be responsible for administrative aspects of the study including but not limited to study initiation, management of SAE reports, monitoring, and data management.

11.2.2. Interactive Response Technology

IRT will be used in this study. Before participants are screened or enrolled, the IRT vendor will provide each study site with the necessary training, a user manual, and access rights to the system.

11.2.3. Electronic or Remote Data Capture

Participant information will be captured and managed by study sites on eCRFs by a Web-based electronic data capture tool configured by the Sponsor or a CRO and hosted by the electronic/remote data capture vendor.

11.2.4. Bioanalytical Laboratory for Laboratory Assessments

A designated laboratory will be selected by the Sponsor to analyze PK and PD samples collected for this study.

11.2.5. Central Facility for Other Assessments

A central facility has been selected by the Sponsor to read and interpret central reads of ECGs for this study. Similarly, a central facility has been selected by the Sponsor to read and interpret all imaging data for efficacy and safety evaluations. This central facility will NOT be used to screen participant imaging data for enrollment purposes.

11.2.6. Central Review of Raters

The Sponsor has selected a rater management group to establish rater qualifications, provide study-specific training about the rater process, and provide oversight. As part of the oversight process, the rater management group will incorporate a central review of the raters to ensure that data are consistently rated across sites.

11.3. Study Committees

11.3.1. Independent Data Monitoring Committee

An IDMC will be formed to review ongoing safety data and tolerability data. Members of the IDMC will not be allowed to participate as Investigators in this study. The IDMC will review safety data on an ongoing basis to ensure the safety and welfare of participants.

The IDMC will include a clinical chair, a statistician, and physician(s) experienced in clinical studies and stroke, who are not participating in this study as an investigator. One of the members will be a nephrologist with extensive experience in acute kidney injury. Details of the composition, roles, responsibilities, and processes of the IDMC will be documented in the committee charter. The frequency and extent of the unblinded and/or blinded safety reviews will be outlined in the committee charter.

In addition to routine independent safety review, the IDMC will be tasked to undertake the following activities:

- Real time assessment of incident ICH cases with routine and ad hoc reviews;
- Adjust the intensity, frequency, and scope of review as cases accrue and at the planned review(s) and interim analysis;
- Help determine criteria for dose selection and make recommendations to the Sponsor regarding the dose to advance to Part 2 of the study; and
- Following the review of the first 100 participants, help adjust the intensity, frequency, and scope of renal testing and monitoring.

12. ADMINISTRATIVE PROCEDURES

12.1. Study Site Initiation

The Investigator must not screen any participants prior to the Sponsor completing a study initiation visit. This initiation visit with the Investigator and other site staff, as appropriate, will include a detailed review of the protocol, study procedures, and study responsibilities.

12.2. Quality Control and Quality Assurance

Quality control procedures will be implemented at each stage of data handling to ensure that all data are reliable and have been processed correctly. Data anomalies will be communicated to the sites for clarification and resolution, as appropriate. The Investigator is responsible for endorsing all eCRF data prior to any interim or final database lock.

During and/or after completion of the study, quality assurance officers named by the Sponsor, or the regulatory authorities may wish to perform onsite audits or inspections. The Investigator will be expected to cooperate with any audit or inspection and to provide assistance and documentation (including source data) as requested.

12.3. Monitoring of the Study

The Investigator must permit study-related monitoring by providing direct access to source data and to the participants' medical histories. Source data must be attributable, legible, contemporaneous, original, accurate, and complete. Changes to source data must be traceable, not obscure the original entry, and be explained if necessary (e.g., with an audit trail). The Investigator should maintain a record of the location(s) of essential documents.

The Medical Monitor will visit the study site at regular intervals during the study and after the study has completed, as appropriate. A clinical site monitoring plan will detail who performs the monitoring, how often, and the extent of review. It also will provide the monitoring strategy, with emphasis on participant safety, data integrity, and critical data and processes.

During these visits, CRFs, supporting documentation, and essential documentation related to the study will be reviewed and any discrepancies or omissions will be resolved. Documentation of results will be provided to the Sponsor in a timely fashion to allow follow-up and verification of compliance with the monitoring plan. Remote evaluation of data (centralized monitoring) may also be conducted and reported as defined in the monitoring plan.

Monitoring visits must be conducted according to the applicable ICH and GCP guidelines to ensure the protection of participant rights and well-being, protocol adherence, quality of data (accurate, complete, and verifiable), study treatment accountability, compliance with regulatory requirements, and continued adequacy of the investigational site and its facilities.

12.4. Study Funding

The Sponsor is responsible for setting up funding of the study. All financial details are provided in the separate contracts between the institution, Investigator, and the Sponsor.

12.5. Publications

Details are included in the clinical trial agreement for this study.

13. REFERENCES

- ACTIVASE (alteplase) for injection Package Insert. South San Francisco, CA: Genentech, Inc.
- Adams HP, Bendixen BH, Kappelle LJ, et al. Classification of subtype of acute ischemic stroke. Definitions for use in a multicenter clinical trial. TOAST. Trial of Org 10172 in Acute Stroke Treatment. *Stroke*. 1993;24(1):35-41.
- Albers GW, Marks MP, Kemp S, et al. Thrombectomy for Stroke at 6 to 16 Hours with Selection by Perfusion Imaging. *N Engl J Med*. 2018 Feb 22;378(8):708-718.
- Albers GW, Juma M, Purdon B, et al. Tenecteplase for Stroke at 4.5 to 24 Hours with Perfusion-Imaging Selection. *N Engl J Med*. 2024;390(8):701-711.
- Berge E, Whiteley W, Audebert H, et al. European Stroke Organisation (ESO) guidelines on intravenous thrombolysis for acute ischaemic stroke. *Eur Stroke J*. 2021;6(1):I-LXII.
- Clarençon F, Durand-Zaleski I, Premat K, et al. Evaluation of mechanical thrombectomy in acute ischemic stroke related to a distal arterial occlusion: A randomized controlled trial. *Int J Stroke*. 2024;19(3):367-372.
- Dhand A, Reeves MJ, Mu Y, et al. Mapping the Ecological Terrain of Stroke Prehospital Delay: A Nationwide Registry Study. *Stroke*. 2024;55(6):1507-1516.
- GBD 2019 Stroke Collaborators. Global, regional, and national burden of stroke and its risk factors, 1990-2019: a systematic analysis for the Global Burden of Disease Study 2019. *Lancet Neurol*. 2021;20(10):795-820.
- Goyal M, Ospel JM, Ganesh A, et al. ESCAPE-MeVO Investigators. Endovascular Treatment of Stroke Due to Medium-Vessel Occlusion. *N Engl J Med*. 2025 Apr 10;392(14):1385-1395.
- Hacke W, Kaste M, Bluhmki E, et al. Thrombolysis with alteplase 3 to 4.5 hours after acute ischemic stroke. *N Engl J Med*. 2008;359(13):1317-29.
- Hasumi K, Suzuki E. Impact of SMTP Targeting Plasminogen and Soluble Epoxide Hydrolase on Thrombolysis, Inflammation, and Ischemic Stroke. *Int J Mol Sci*. 2021;22(2);954.
- Hollist M, Morgan L, Cabatbat R, Au K, Kirmani MF, Kirmani BF. Acute Stroke Management: Overview and Recent Updates. *Aging Dis*. 2021;12(4):1000-1009.
- Kleindorfer DO, Towfighi A, Chaturvedi S, et al. 2021 Guideline for the Prevention of Stroke in Patients With Stroke and Transient Ischemic Attack: A Guideline From the American Heart Association/American Stroke Association. *Stroke*. 2021;52(7):e364-e467.
- Li XY, Kong XM, Yang CH, et al. Global, regional, and national burden of ischemic stroke, 1990-2021: an analysis of data from the global burden of disease study 2021. *EClinicalMedicine*. 2024;75:102758.
- Mac Grory B, Xian, Y., et al. Exploring the unmet need in acute ischemic stroke patients not treated with intravenous alteplase: The get with the guidelines-stroke registry. *Stroke: Vascular and Interventional Neurology*. 2021;2:1-13.

- Martin SS, Aday AW, Almarzooq ZI, et al. American Heart Association Council on Epidemiology and Prevention Statistics Committee and Stroke Statistics Subcommittee. 2024 Heart Disease and Stroke Statistics: A Report of US and Global Data from the American Heart Association. *Circulation*. 2024;149(8):e347-e913.
- Menon BK, Buck BH, Singh N, et al. AcT Trial Investigators. Intravenous tenecteplase compared with alteplase for acute ischaemic stroke in Canada (AcT): a pragmatic, multicentre, open-label, registry-linked, randomised, controlled, non-inferiority trial. *Lancet*. 2022 Jul 16;400(10347):161-169.
- Meschia JF, Bushnell C, Boden-Albala B, et al. Guidelines for the primary prevention of stroke: a statement for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke*. 2014;45(12):3754-832.
- National Institute of Neurological Disorders and Stroke rt-PA Stroke Study Group. Tissue plasminogen activator for acute ischemic stroke. *N Engl J Med*. 1995 Dec 14;333(24):1581-7.
- Nogueira RG, Jadhav AP, Haussen DC, et al. Thrombectomy 6 to 24 Hours after Stroke with a Mismatch between Deficit and Infarct. *N Engl J Med*. 2018 Jan 4;378(1):11-21.
- Powers WJ, Rabinstein AA, Ackerson T, et al. Guidelines for the Early Management of Patients With Acute Ischemic Stroke: 2019 Update to the 2018 Guidelines for the Early Management of Acute Ischemic Stroke: A Guideline for Healthcare Professionals From the American Heart Association/American Stroke Association. *Stroke*. 2019;50(12):e344-e418.
- Psychogios M, Brehm A, Ribo M, et al. DISTAL Investigators. Endovascular Treatment for Stroke Due to Occlusion of Medium or Distal Vessels. *N Engl J Med*. 2025 Apr 10;392(14):1374-1384.
- Rai AT, Seldon AE, Boo S, et al. A population-based incidence of acute large vessel occlusions and thrombectomy eligible patients indicates significant potential for growth of endovascular stroke therapy in the USA. *J Neurointerv Surg*. 2017 Aug;9(8):722-726.
- Rao NM, Levine SR, Gornbein JA, et al. Defining clinically relevant cerebral hemorrhage after thrombolytic therapy for stroke: analysis of the National Institute of Neurological Disorders and Stroke tissue-type plasminogen activator trials. *Stroke*. 2014;45(9):2728-33.
- Rodeghiero F, Tosetto A, Abshire T, et al. ISTH/SSC bleeding assessment tool: a standardized questionnaire and a proposal for a new bleeding score for inherited bleeding disorders. *J Thromb Haemost*. 2010;8(9):2063-5.
- Sarraj A, Hassan AE, Abraham MG, et al. Trial of Endovascular Thrombectomy for Large Ischemic Strokes. *N Engl J Med*. 2023 Apr 6;388(14):1259-1271.
- Sawada H, Nishimura N, Suzuki E, et al. SMTP-7, a novel small-molecule thrombolytic for ischemic stroke: a study in rodents and primates. *J Cereb Blood Flow Metab*. 2014;34(2):235-41.
- Seners P, Yuen N, Mlynash M, et al. Quantification of Penumbra Volume in Association With Time From Stroke Onset in Acute Ischemic Stroke With Large Vessel Occlusion. *JAMA Neurol*. 2023;80(5):523-528.

Tsao CW, Aday AW, Almarzooq ZI, et al. Heart Disease and Stroke Statistics-2022 Update: A Report From the American Heart Association. *Circulation*. 2022;145(8):e153-e639.

Turc G, Bhogal P, Fischer U, et al. European Society for Minimally Invasive Neurological Therapy (ESMINT) Guidelines on Mechanical Thrombectomy in Acute Ischemic Stroke. *J Neurointerv Surg*. 2023 Aug;15(8):e8.

Waqas M, Rai AT, Vakharia K, Chin F, Siddiqui AH. Effect of definition and methods on estimates of prevalence of large vessel occlusion in acute ischemic stroke: a systematic review and meta-analysis. *J Neurointerv Surg*. 2020 Mar;12(3):260-265.

14. SIGNED AGREEMENT OF THE STUDY PROTOCOL

I have read the foregoing protocol, Optimizing Reperfusion to Improve Outcomes and Neurologic Function (ORION): A Multicenter, Double-Blind, Placebo-Controlled, Randomized, Parallel-Group, Phase 2/3 Study to Evaluate the Efficacy and Safety of JX10 in Acute Ischemic Stroke with Late Presentations, and agree to conduct the study according to the protocol and the applicable ICH guidelines and GCP regulations, and to inform all who assist me in the conduct of this study of their responsibilities and obligations.

Investigator's Signature

Date

Investigator's Name (Print)

Study Site (Print)